CS2 AI COACHING SYSTEM

Ultimate CS2 Coach

Part 1A — The Brain

Neural architecture, training, JEPA, VL-JEPA, Superposition Layer, LSTM + Mixture of Experts

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Ultimate CS2 Coach — Part 1A: The Brain

Topics: Complete neural architecture, training regime, deep learning models (JEPA, VL-JEPA, LSTM+MoE), Coach Introspection Observatory, 25-dim contract, NO-WALLHACK principle, and system architecture overview (startup, Qt/PySide6 desktop interface, quad-daemon architecture, general pipeline).

Author: Renan Augusto Macena

Introduction by Renan

This project is the result of countless hours of preparation, refinement, and above all, research. Counter-Strike is my football, something I have been passionate about my whole life, like music. By now I have more than 10 thousand hours in this game and I have been playing since 2004, I have always wanted a professional guide, like that of real professional players, to understand what it truly looks like when someone trains the right way, and plays the right way knowing what is right or wrong and not just guessing it. At this point my confidence is tied to my knowledge of the game, but it is still far from the level I would like to play at. I imagine many people like me, in this game or perhaps in others, feel the same way: they know they have experience and knowledge of what they are doing, but they realize that professional players are so much more skilled and refined that even years of experience cannot match a fraction of what these professionals offer.

This project attempts to "bring to life", using advanced techniques, a definitive coach for Counter-Strike, one that understands the game, evaluates dozens of aspects with clarity and refined judgment, absorbs and becomes smarter over time.. My ultimate goal is to have it ingest, process and learn from all pro matches ever, not all playoffs, besides being unrealistic for the size and processing time, it would be useless since my goal is to have this definitive coach based on the best of the best. The model for the "low-skill" player will always be the user, he or she will never have demos and matches from which this coach can learn; instead, this is a completely different approach method, since the coach will use its high-quality model to compare with the user's model, showing how far the user really is from the level of a professional player and, more importantly, adapting the teachings to help the user reach the pro level, adapting to each different play style: if I am an awper, slowly but surely this coach will teach me how to become a pro awper, and this applies to every role in the game in which a user wants to become a professional.

And now with the passion I am developing for programming, understanding all this complicated stuff about database management, SQL, machine learning models, Python, and how elegant it can be, from all these technical steps of implementation, tuning, and API creation, and understanding what that .dll is that I have seen my whole life inside game folders, understanding what those frameworks like Kivy are for to create the graphics, learning what it really means to create software, I had to learn to go on Linux (I am definitely not a professional, and if I had not followed the same instructions I created while researching and continuing to work on this project, I would not have made it) to understand how to build the program, how to make it cross-platform. From Linux, I built the APK version (I still have to finish working on that one too), and at least understood the most important concepts of Compilation, then I finished the desktop build on Windows and I did everything else, compilation and packaging. In short, pushing yourself to the limit to understand, challenging yourself so much, on so many different aspects that you have to either absorb not just something but also the specifications and the complex picture of everything, or you will get nowhere.

I used tools, like Claude CLI on the windows and linux terminal, to help me organize the code and fix syntax errors. I created a folder in the project folder containing a "Tool Suite" made up of Python scripts that I helped create and that have a ton of useful functions, from debugging to modifying values, rules, functions, strings at the global or local level of the project so that from a single input inside that tool, I can change whatever I want "anywhere", even change things on the graphical dashboard with inputs. I have not been sleeping for about twenty consecutive days (since December 24th, 2025), yes, but I think it is worth it, and I know well that this is only a training in the end that I figured out how to do on my own from beginning to end, so there are certainly mistakes around. Whether it can really be useful, really functional, etc., etc., and many other nice things, I do not know. I do not want to overstate what I am doing. I am doing it, but I have really put a lot of effort into it.

I hope something in there can be useful.

> *Note: The UI framework was later migrated from Kivy to Qt/PySide6 (March 2026), as documented in Part 3.*

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Part 1A — The Brain: Neural Architecture and Training (this document)

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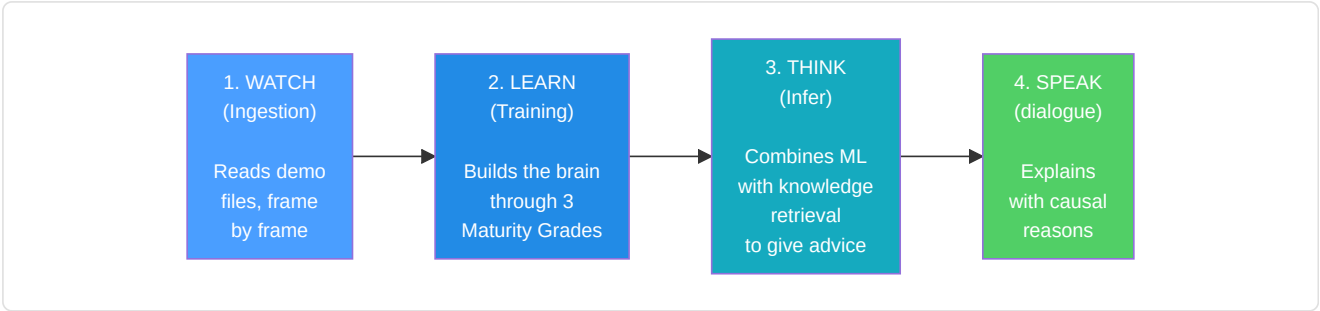
1. Executive Summary

CS2 Ultimate is a **hybrid AI-based coaching system** for Counter-Strike 2 (CS2). It combines deep learning models (JEPA, VL-JEPA, LSTM+MoE, a 7-component RAP architecture (Perception, LTC+Hopfield Memory, Strategy, Pedagogy, Causal Attribution, Positioning Head + external Communication), a Neural Role Classification Head), a Coach Introspection Observatory, a Retrieval-Augmented Generation (RAG), the COPER experience bank, game-theoretic search, Bayesian belief modeling, and a Quad-Daemon architecture (Hunter, Digester, Teacher, Pulse) into a unified pipeline that:

1. **Ingests** professional and user demo files, extracting tick-level and match-level state statistics.
2. **Trains** multiple neural network models through a phased program with maturity gates (3 levels: CALIBRATION → LEARNING → MATURE). 3. **Infers** training advice by fusing machine learning predictions with tactical knowledge retrieved semantically via a 4-tier fallback chain (COPER → Hybrid → RAG → Baseline).
3. **Explains** its reasoning through causal attribution, template-based narratives, professional player comparisons, and an optional LLM refinement (Ollama).

The system contains **≈ 103,600 lines of Python** spread across 411 `.py` files under `Programma_CS2_RENAN/`, extending over **eight logical AI subsystems** (NN Core with VL-JEPA, RAP Coach + RAP Lite, Coaching Services, Knowledge & Retrieval, Analysis Engines (11), Processing & Feature Engineering, Data Sources, Coaching Engines), a training Observatory, a Control module (Console with REST API, DB Governor, Ingest Manager, ML Controller), a Quad-Daemon architecture for background

automation (Hunter, Digester, Teacher, Pulse), a Qt/PySide6 desktop interface with 15 screens and MVVM patterns (migrated from Kivy in March 2026), a complete ingestion system (with a dedicated HLTV subsystem: HLTVApiService, CircuitBreaker, RateLimiter), storage and reporting, a **Tools Suite** with 41 Python validation and diagnostic scripts (29 root + 12 in the package — Goliath Hospital, headless validator, Ultimate ML Coach Debugger, validate_coaching_pipeline, ingest_pro_demos, dead_code_detector, dev_health, rebuild_monolith, tick_census), a specialized tri-database architecture with 18 SQLModel tables in the monolith (+ 3 in the separate HLTV database + 3 in the per-match databases = 24 total), and a **Test Suite** with 99 test files organized into 6 categories: analysis/theory, coaching/training, ML/models, data/storage, UI/playback, integration/misc. The project has gone through a systematic **13-phase remediation process** that resolved 412+ issues of code quality, ML correctness, security, and architecture, including the elimination of label leakage in training (G-01), implementation of the visual danger zone in the view tensor (G-02), automatic calibration of the Bayesian estimator (G-07), and correction of the COPER coaching fallback (G-08), followed by a **second remediation wave** that resolved an additional 162 issues (31 HIGH + 131 MEDIUM) related to thread safety, schema drift, Qt lifecycle, and observability hardening, and a **third wave** (April 2026) that addressed 40+ additional issues including type safety (SA-14–SA-27), dependency pinning (DEP-1), checkpoint security (CTF-1/2), DataLineage audit trail (DL-1), and UI/UX fixes (UX-1/2/3). The end-to-end pipeline was completed on March 12, 2026: 11 professional demos ingested, 17.3M tick rows, 6.4GB database, pre-trained JEPA (train loss 0.9506, val loss 1.8248). April 2026 update: 156 per-match databases, fully trained AdvancedCoachNN (`latest.pt`), HLTV database populated with **161 real professional players** (32 teams, 156 stat cards), HybridCoachingEngine enhanced with automatic reference pro selection by name in coaching feedback, **Coach Book v4** expanded to **502 entries** of tactical knowledge in 8 JSON files (7 maps + general) with 13 categories, **CoachingDialogueEngine** for multi-turn coaching with per-player and per-round drill-down, **MovementQualityAnalyzer** (11th analysis engine), **EloAugmentedPredictor** for win probability with Elo features, **PlusMinus rating metric**, COPER enhancements (TrueSkill uncertainty, CRUD semantics, prioritized replay), bombsite-relative position encoding, demo prioritization by coaching variance with quality scoring via Huber model, and **CS2 Coach Bench** 200-question evaluation benchmark.



> 406 .py files · 102,000+ lines · 8 AI subsystems + Observatory + Control Module (REST API) + Quad-Daemon + Qt/PySide6 Desktop UI (15 screens) + 41 Tools (29 root + 12 inner) · 94 test files · 21 SQLModel tables (monolith) + 3 per-match · Tri-database architecture (database.db + hltv_metadata.db + per-match database) · FAISS vector indexing (IndexFlatIP 384-dim) · i18n Internationalization (EN/IT/PT) · WCAG 1.4.1 Accessibility (theme.py) · 12 comprehensive audit reports (incl. 140KB literature review, 30 peer-reviewed articles) · 610+ issues resolved (412 in 13 phases + 162 in second wave + 40+ in third wave) · End-to-end pipeline completed (156 per-match DB · JEPA + AdvancedCoachNN trained · 161 real HLTV players, 32 teams, 156 stat cards · Coach Book v4: 502 entries, 8 files, 13 categories · CS2 Coach Bench: 200 questions)

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2. System Architecture Overview

The system is divided into **6 main subsystems** that work together like the departments of a company. Each subsystem has a specific task and the data flows between them in a well-defined pipeline.

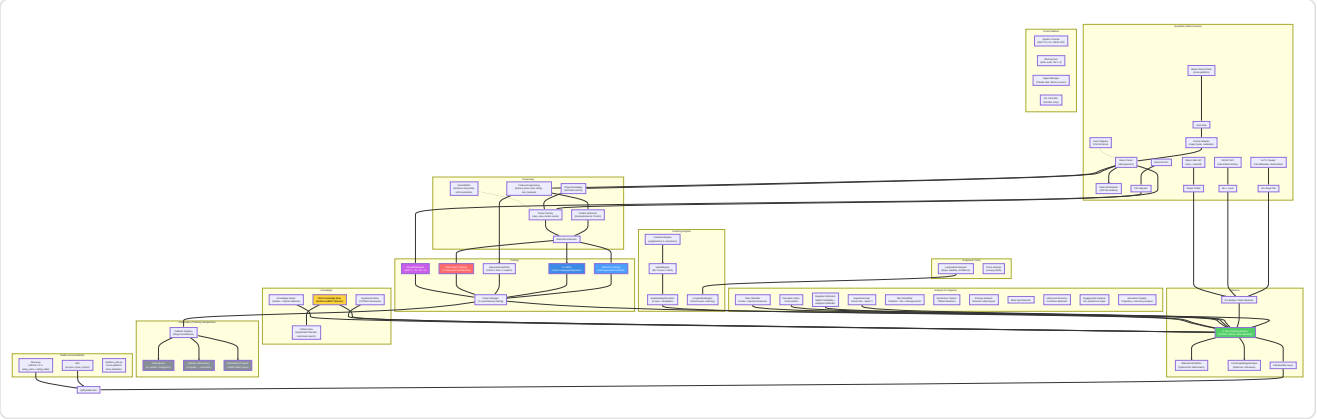


Diagram explanation: This big diagram is like a **treasure map** that shows how information travels through the system. The journey starts top left with the raw game recordings (`.dem` files, HLTV data, CSV files): think of them as **raw ingredients** arriving in the kitchen. These ingredients pass through the Processing section where they are **chopped, measured, and prepared** (features are extracted, vectors are created). Then they reach the Training section where five different "chefs" (JEPA, VL-JEPA, AdvancedCoachNN, RAP, and NeuralRoleHead) each learn their own cooking style. The Observatory is the **quality control inspector** that watches every training session, checking whether the chefs are improving, stalling, or panicking. The Knowledge section is like the **cookbook shelf**: it contains tips (RAG), past cooking successes (COPER), and relationships between ingredients (Knowledge Graph). The Inference section is where the **head chef** combines everything — acquired skills, recipe books, and pro chef techniques — to create the final dish: coaching advice. The Analysis section is like having **food critics** evaluating specific qualities: "Is it too spicy?" (momentum), "Is it creative?" (deception index), "Did they forget an ingredient?" (blind spots). Behind the scenes, the **Quad-Daemon architecture** (Hunter, Digester, Teacher, Pulse) works tirelessly like the automated kitchen staff: it scans new ingredients, prepares them, and updates the chefs' skills without ever stopping. Everything flows down and to the right until it reaches the Qt/PySide6 graphical interface, the **plate** where the user sees the final result.

Data flow summary

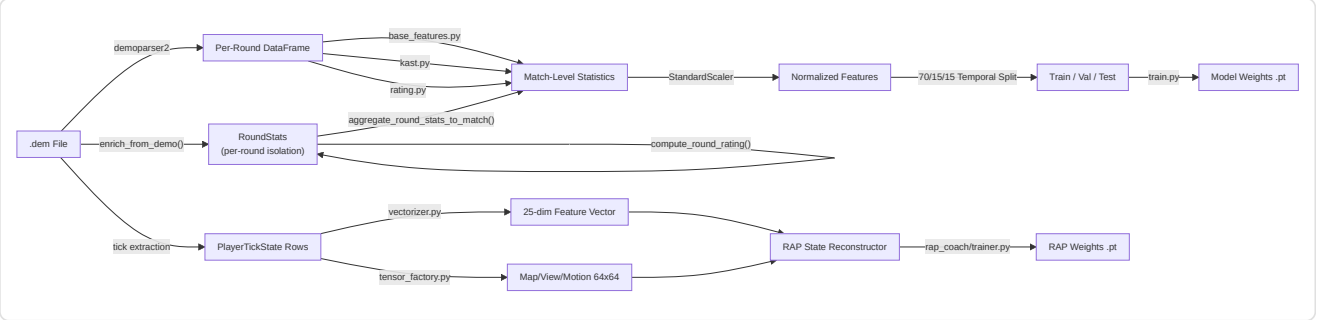
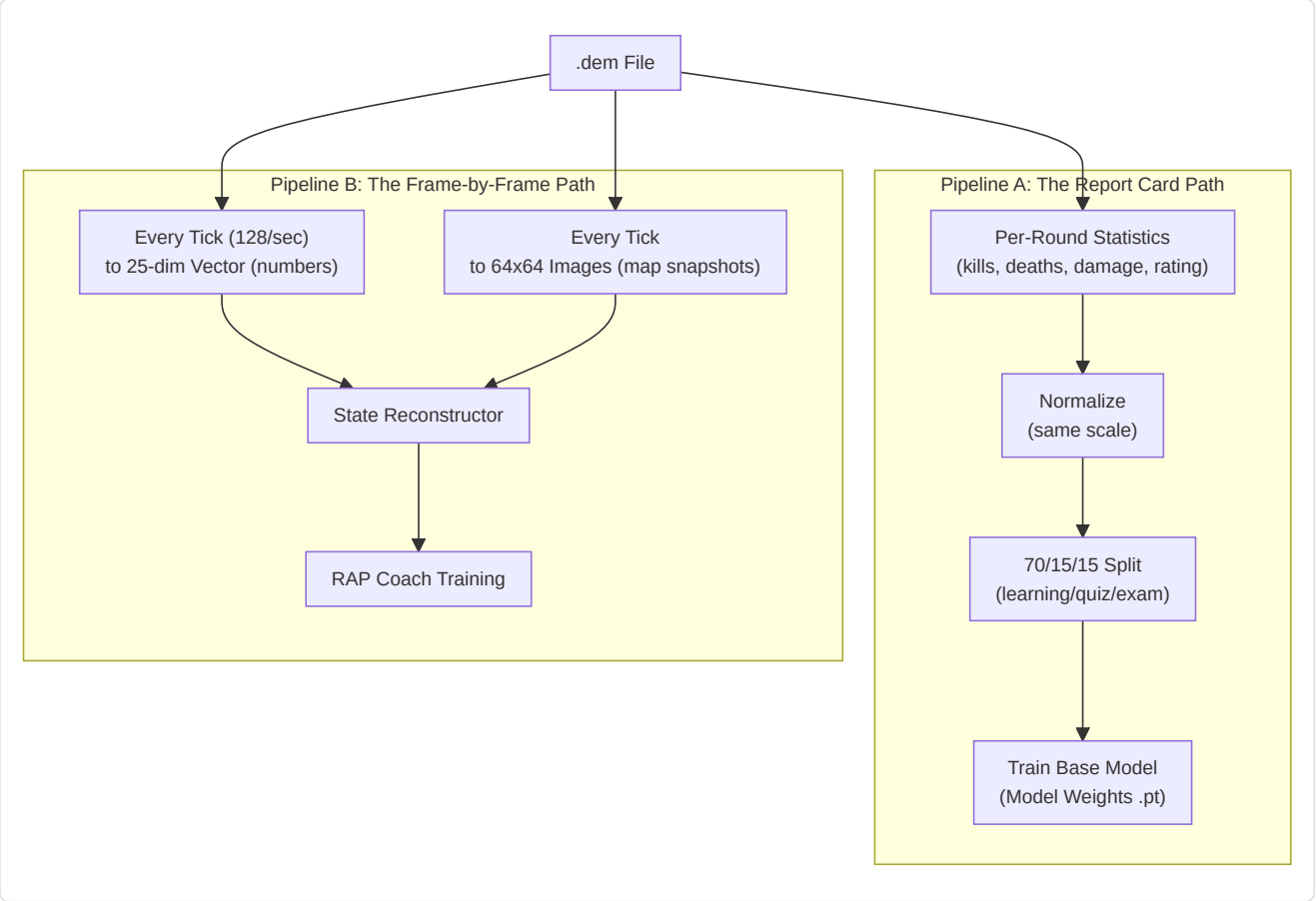


Diagram explanation: This diagram shows the **two parallel assembly lines** inside the processing department. Think of a match recording (`.dem` file) as a **long movie**. The **upper assembly line** watches the movie and writes summary statistics, like a report card for each match (kills, deaths, damage, etc.). These report cards are normalized (put on the same scale, like converting all temperatures to Celsius), split into study groups (70% for learning, 15% for quizzes, 15% for final exams), and used to train the basic training model. The **lower assembly line** is more detailed: it examines the movie **frame by frame** (every "tick" of the game clock), measuring 25 pieces of information about each player at every moment (position, health, what they see, economy, etc.) and creating 64x64 pixel "snapshots" of the map. Both numbers and images are fed into the RAP Coach's State Reconstructor, which combines them into a complete "what was happening at this precise moment" picture — and this is what the advanced RAP Coach model learns from.



NO-WALLHACK Principle and 25-dim Contract

Two fundamental architectural invariants run through the entire system:

1. NO-WALLHACK Principle: The AI coach **sees only what the player legitimately knows**. When the `PlayerKnowledge` module is available, the tensors generated by the `TensorFactory` encode exclusively legitimate information: teammates (always visible), enemies in "last-known" positions (with temporal decay, τ = 2.5s), own and observed utility. No "wallhack" information (real enemy positions that are not visible) ever enters the perception system. When `PlayerKnowledge` is `None`, the system falls back to a legacy mode with simplified tensors.

2. 25-dim Contract (`FeatureExtractor`): The `FeatureExtractor` in `vectorizer.py` defines the canonical 25-dimensional feature vector (`METADATA_DIM = 25`) used by **all** models (AdvancedCoachNN, JEPA, VL-JEPA, RAP Coach) both in training and inference. Any change to the feature vector happens **exclusively** in the `FeatureExtractor` — no other module may define its own features. This guarantees end-to-end dimensional consistency.

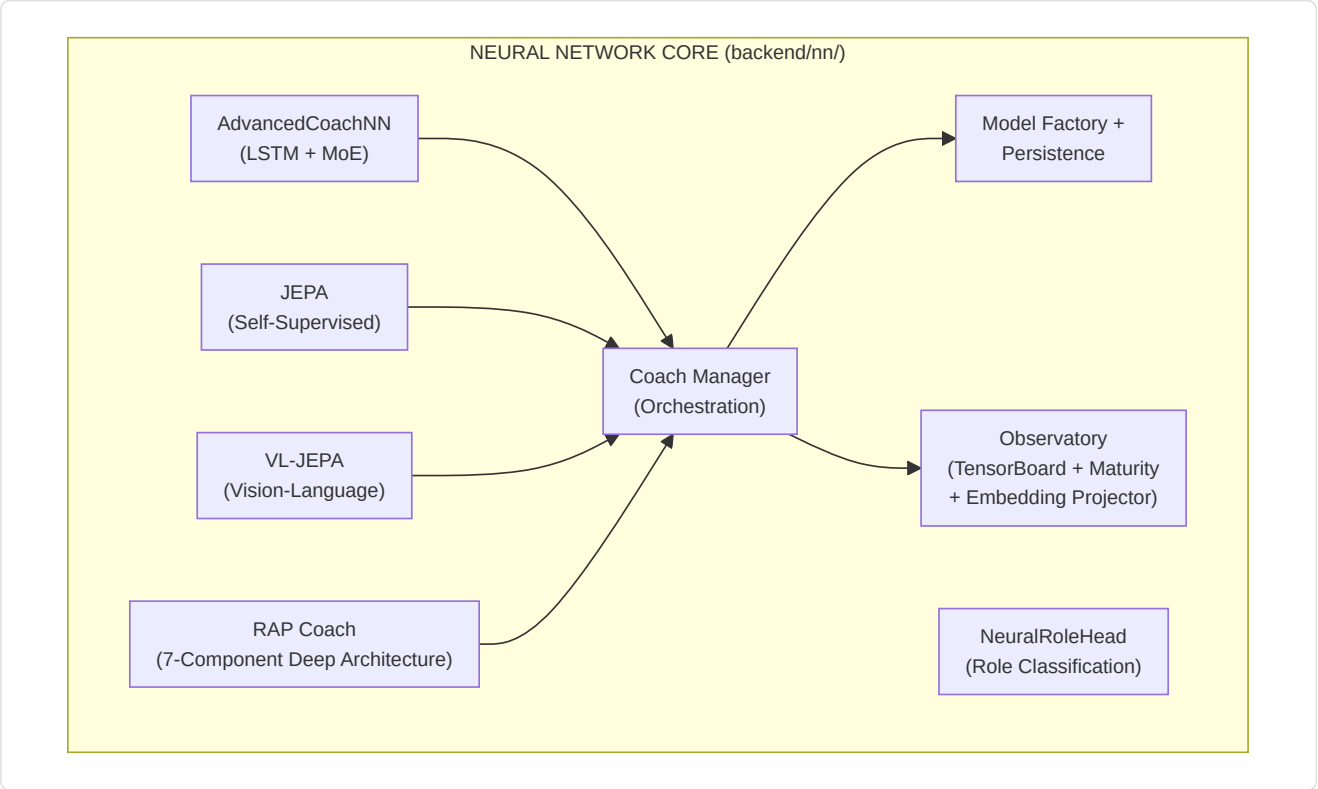
```
 0: health/100      1: armor/100       2: has_helmet      3: has_defuser
 4: equip/10000     5: is_crouching    6: is_scoped       7: is_blinded
 8: enemies_vis     9: pos_x/4096     10: pos_y/4096     11: pos_z/1024
12: view_x_sin     13: view_x_cos     14: view_y/90      15: z_penalty
16: kast_est       17: map_id         18: round_phase
19: weapon_class   20: time_in_round/115  21: bomb_planted
22: teammates_alive/4  23: enemies_alive/5  24: team_economy/16000

Position normalization: `np.clip(pos_x / cfg.pos_xy_extent, -1.0, 1.0)` where
`pos_xy_extent=4096.0` (default, configurable per map). The clip to [-1,1] protects
against out-of-range coordinates that would produce features > 1.0 in non-normalized
modules.
```

---

3. Subsystem 1 — Neural Network Core

Program folder: `backend/nn/` **Key files:** `model.py`, `jepa_model.py`, `jepa_train.py`, `jepa_trainer.py`, `coach_manager.py`, `training_orchestrator.py`, `config.py`, `factory.py`, `persistence.py`, `role_head.py`, `training_callbacks.py`, `tensorboard_callback.py`, `maturity_observatory.py`, `embedding_projector.py`, `dataset.py`, `data_quality.py`, `evaluate.py`, `train_pipeline.py`, `training_monitor.py`, `early_stopping.py`, `ema.py`, `training_controller.py`, `win_probability_trainer.py`, `training_config.py`

This subsystem contains all the neural network models, the "brain" of the coaching system. It includes six distinct model architectures (AdvancedCoachNN, JEPA, VL-JEPA, RAP Coach, RAP Lite, NeuralRoleHead), a training manager, a Coach Introspection Observatory, and utilities for model creation and persistence.



-AdvancedCoachNN (LSTM + Mixture of Experts)

Defined in `model.py`, this is the foundation of supervised coaching.

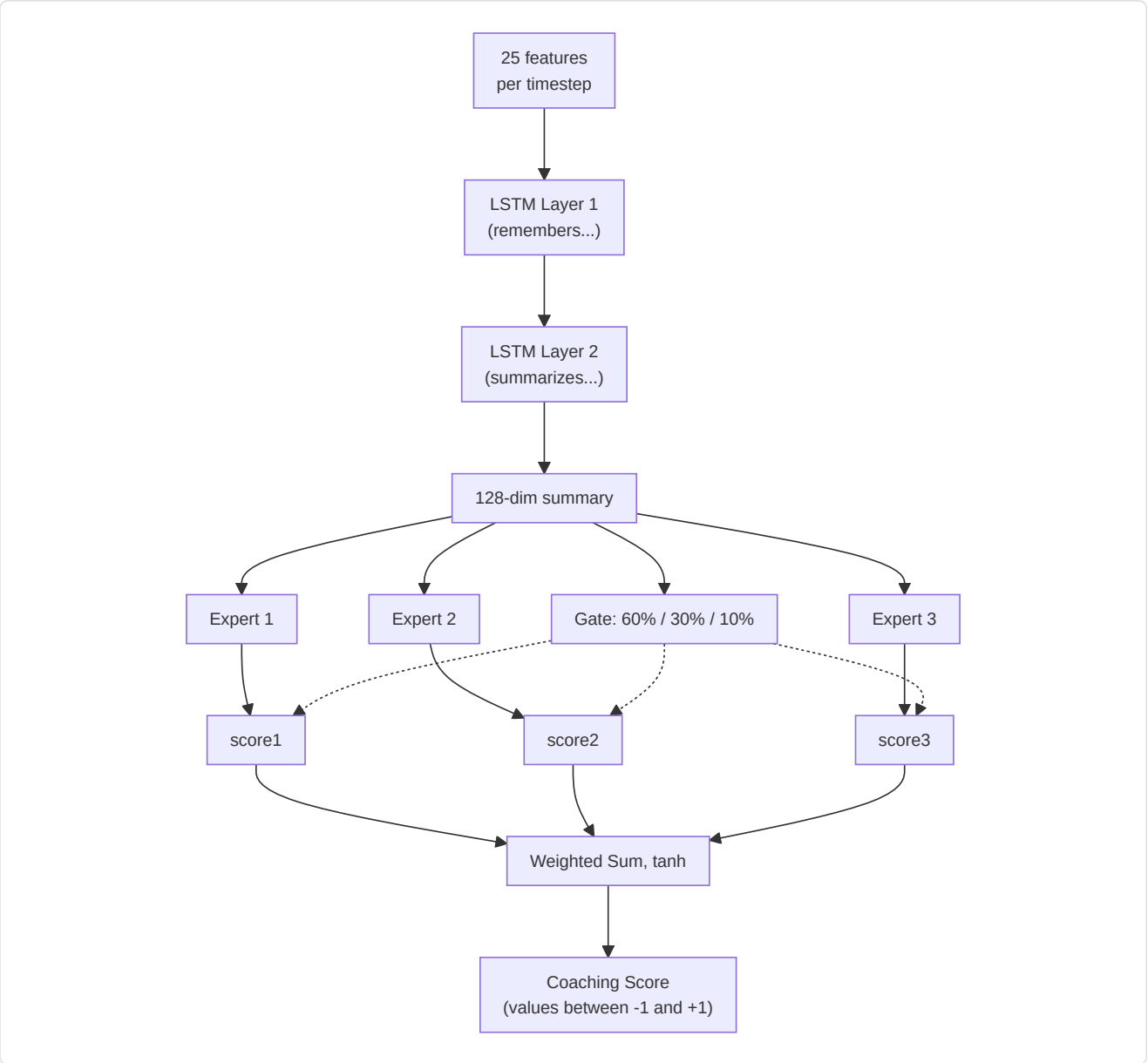
Component	Detail
Input dimension	25 features (`METADATA_DIM` from vectorizer.py)
Config	`CoachNNConfig` dataclass: `input_dim=25`, `output_dim=METADATA_DIM` (default 25, but overridden to `OUTPUT_DIM=10` by `ModelFactory`), `hidden_dim=128`, `num_experts=3`, `num_lstm_layers=2`, `dropout=0.2`, `use_layer_norm=True`
Hidden layers	2-layer LSTM (128 hidden, `batch_first=True`, dropout=0.2) with `LayerNorm` post-LSTM
Expert head	3 parallel linear experts (configurable), softmax-gated via a learned gating network
Output	Weighted sum of expert outputs → 10-dimensional coaching score vector. The `CoachNNConfig` defines `output_dim=METADATA_DIM` (25) as default, but `ModelFactory` overrides with `OUTPUT_DIM=10` in production. The alias `TeacherRefinementNN = AdvancedCoachNN` is kept for backward compatibility
Role bias	Optional `role_id` parameter: `gate_weights = (gate_weights + role_bias) / 2.0` — biases expert selection toward role-specific knowledge
Input validation	`_validate_input_dim()` automatically reshapes 1D → `unsqueeze(0).unsqueeze(0)` and 2D → `unsqueeze(0)` for robustness

Each expert module in AdvancedCoachNN: `Linear(128→128) → LayerNorm(128) → ReLU → Linear(128→output_dim)`.

> **Note:** JEPA's `_create_expert()` omits LayerNorm — only `Linear → ReLU → Linear`. This is a deliberate design choice: JEPA experts operate on already normalized latent embeddings, while AdvancedCoachNN experts process raw LSTM outputs that benefit from per-expert normalization.

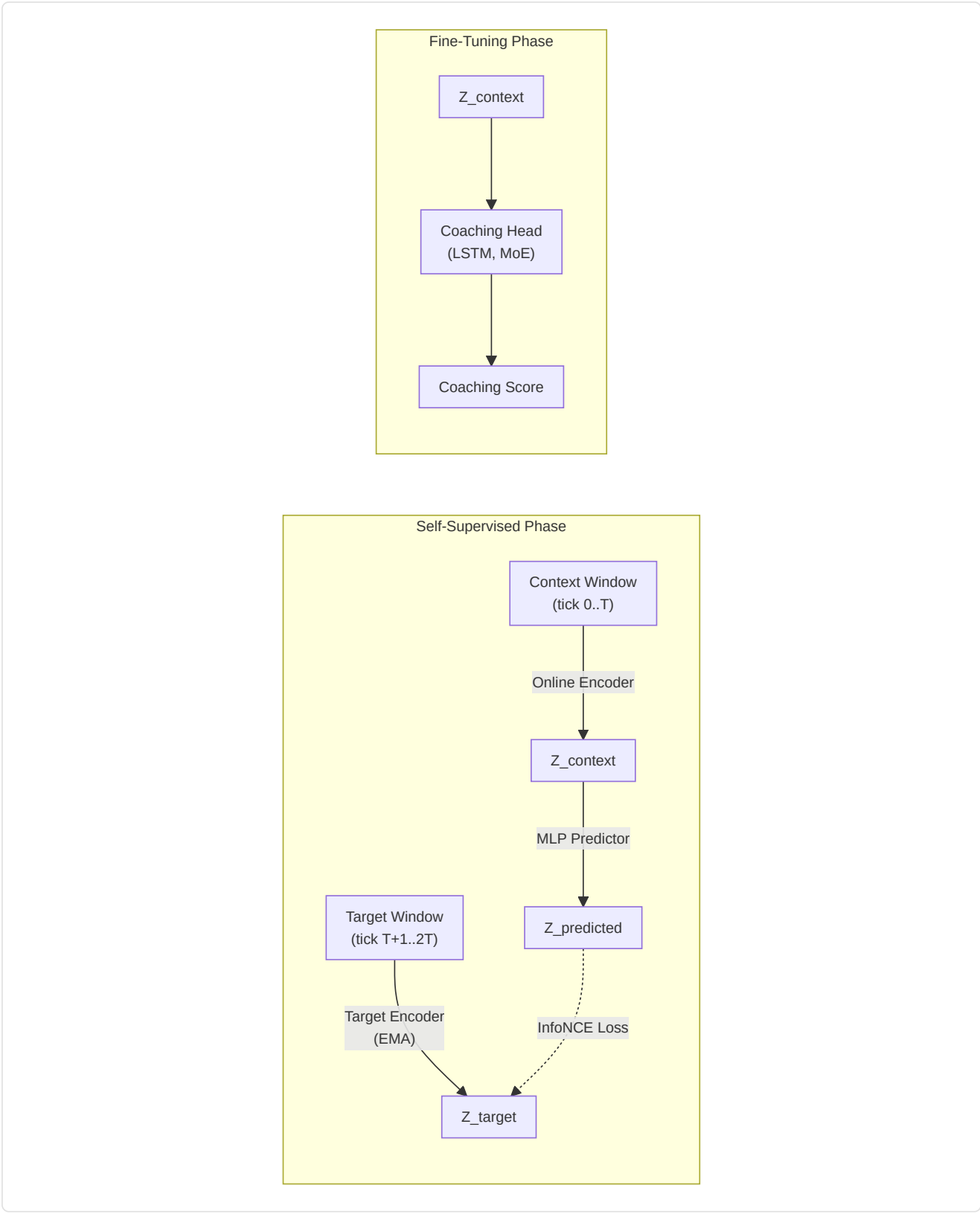
Forward pass (pseudo forward pass):

```
h, _ = LSTM(x) # x: [batch, seq_len, 25]
h = LayerNorm(h[:, -1, :]) # take the last timestep → [batch, 128]
gate_weights = softmax(W_gate · h) # [batch, 3]
expert_outputs = [E_i(h) for i in 1..3]
output = tanh(Σ gate_weights_i × expert_outputs_i)
```



-JEPA Coaching Model (Joint Embedding Predictive Architecture)

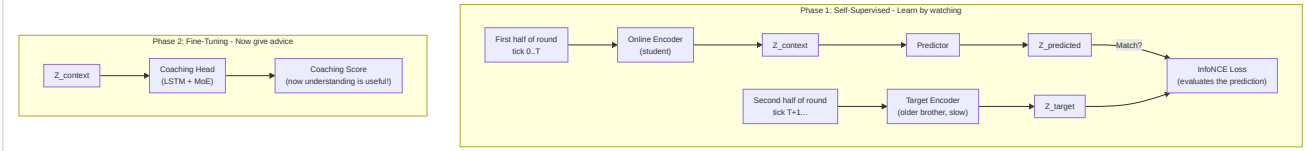
Defined in `jepa_model.py`. A **self-supervised pre-training model** inspired by Yann LeCun's I-JEPA, adapted for sequential CS2 data.


Fine-Tuning Phase
Z_context
Coaching Head
(LSTM, MoE)
Coaching Score
Self-Supervised Phase
Context Window
(tick 0..T)
Online Encoder
Z_context
MLP Predictor
Target Window
(tick T+1..2T)
Z_predicted
Target Encoder
(EMA)
InfoNCE Loss
Z_target

> **Diagram explanation:** The self-supervised phase works like this: imagine watching a movie and pressing pause halfway through a scene. The **Online Encoder** looks at the first half and creates a summary ("here is what I have understood so far"). The **Target Encoder** (a slightly older copy of the same brain, slowly updated) looks at the second half and creates its own summary. Then a **Predictor** tries to guess the summary of the second half using only the summary of the first half. The **InfoNCE Loss** is like a teacher checking: "Does your prediction match what actually happened? And is it different enough from random guesses?". In the Fine-Tuning phase, once the model has acquired good predictive capability, we add a **Coaching Head** on top: now the understanding gained by watching can be used to provide actual coaching scores.

Architecture details:

Module	Parameters
Online encoder	Linear(input_dim, 512) → LayerNorm → GELU → Dropout(0.1) → Linear(512, latent_dim=256) → LayerNorm
Target encoder	Structurally identical; updated via exponential moving average (τ = 0.996). `EMA.state_dict()` returns **cloned** tensors to prevent aliasing (a previous bug allowed accidental modification of target weights through shared references)
Predictor	Linear(256, 512) → LayerNorm → GELU → Dropout(0.1) → Linear(512, 256)
Coaching Head	LSTM(256, hidden_dim, 2 layers, dropout=0.2) → 3 MoE experts → gated output



Pre-training procedure (`jepa_trainer.py`):

1. Loads `PlayerTickState` sequences from professional demo SQLite files.
2. Splits each sequence into context and target windows.
3. Encodes the context via the online encoder + predictor, encodes the target via the target encoder (EMA).
4. Minimizes the **InfoNCE contrastive loss** using in-batch negatives with cosine similarity and temperature τ=0.07.
5. After each batch, performs the EMA update: `θ_target ← τ·θ_target + (1−τ)·θ_online` .
6. **Drift monitoring**: Tracks DriftReport objects; triggers automatic retraining if drift > 2.5σ.
7. **Outcome-based labels (Fix G-01):** The `ConceptLabeler` in VL-JEPA training now generates labels from `RoundStats` data (per-round outcomes: kills, deaths, damage, survival) instead of tick-level features. This eliminates **label leakage** — the previous problem where concept labels were derived from the same features used as input, allowing the model to "cheat" during training without actually learning the patterns. The `label_from_round_stats(rs)` method produces a vector of 16 concept labels based on measurable outcomes. If `RoundStats` data is not available, the system falls back to the legacy heuristic with a one-time log warning.

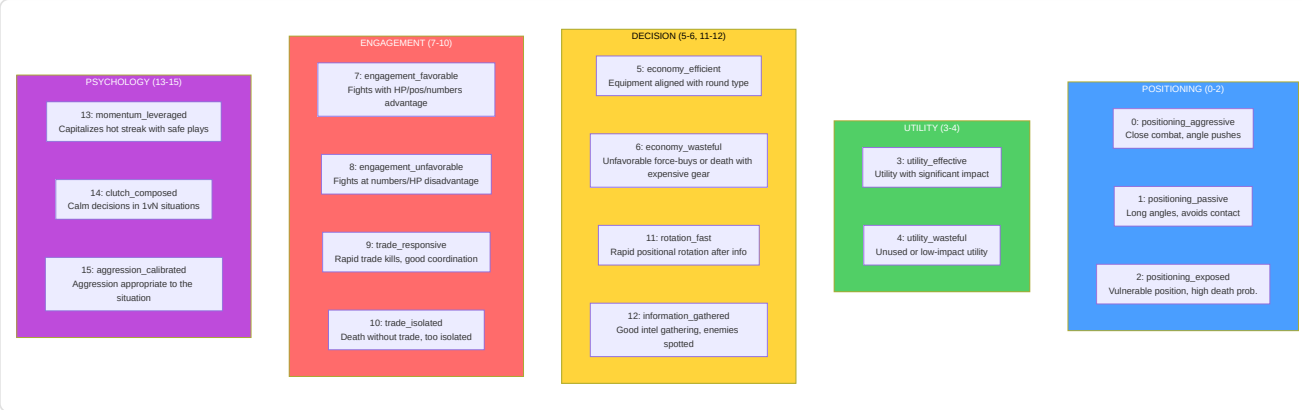
Selective Decoding (`forward_selective`): skips the entire forward pass if the cosine distance between the current and previous embedding is below a threshold (`skip_threshold=0.05`). Uses `1.0 - F.cosine_similarity()` as distance metric and, during the skip operation, returns the previously cached output. This allows efficient real-time inference with dynamic frame skipping: during static gameplay moments (players holding angles), most frames are skipped entirely.

-VL-JEPA: Vision-Language Alignment Architecture with Coaching Concepts

Defined in the second half of `jepa_model.py` . VL-JEPA (**Vision-Language JEPA**) is a **foundational extension** of the JEPACoachingModel that adds an **alignment mechanism between latent embeddings and interpretable coaching concepts**. Inspired by Meta FAIR's VL-JEPA (2026), it maps latent representations into a structured concept space with 16 predefined coaching concepts.

Taxonomy of the 16 Coaching Concepts

The system defines `NUM_COACHING_CONCEPTS = 16` concepts organized into **5 tactical dimensions**:



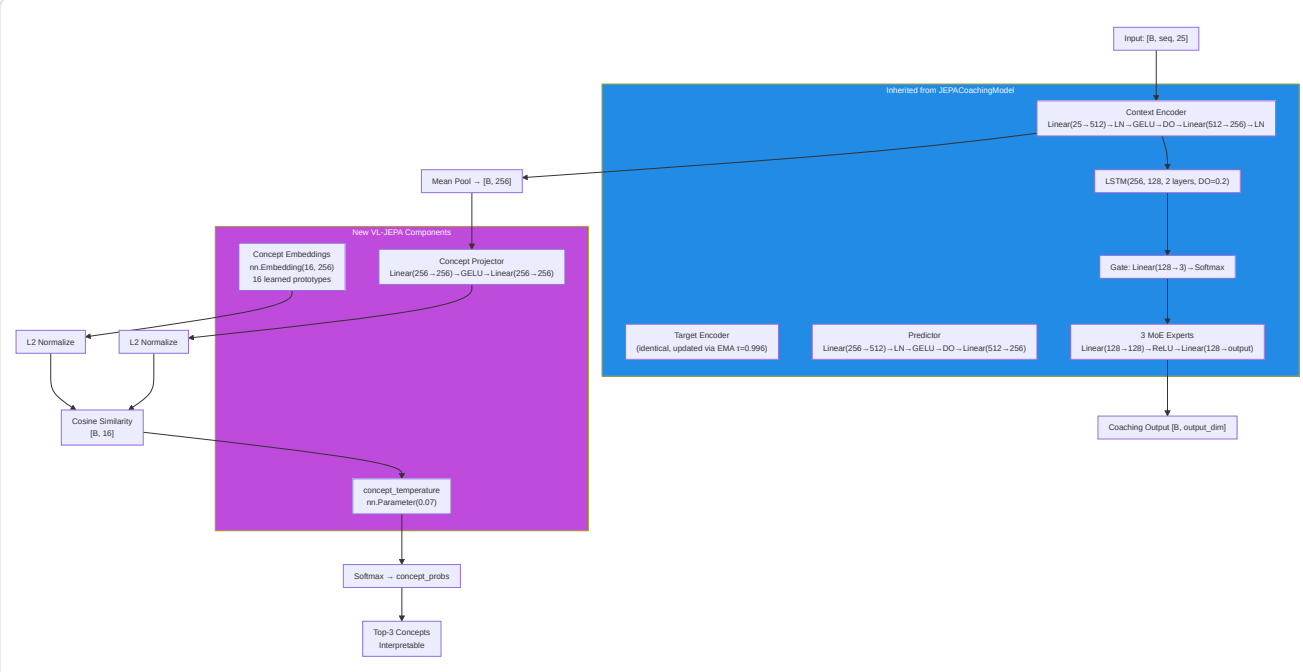
Each concept is defined as an immutable `CoachingConcept` dataclass with `(id, name, dimension, description)`. The global list `COACHING_CONCEPTS` and `CONCEPT_NAMES` are the sources of truth for the whole system.

VLJEPACoachingModel Architecture

`VLJEPACoachingModel` inherits from `JEPACoachingModel` and adds 3 components:

Component	Parameters	Purpose
concept_embeddings	`nn.Embedding(16, latent_dim=256)`	16 concept prototypes learned in latent space
concept_projector	`Linear(256→256) → GELU → Linear(256→256)`	Projects encoder embedding into concept-aligned space
concept_temperature	`nn.Parameter(0.07)`, clamped `[0.01, 1.0]`	Learned temperature for cosine similarity scaling

All parent forward paths (`forward` , `forward_coaching` , `forward_selective` , `forward_jepa_pretrain`) are **preserved unchanged** via inheritance. The new `forward_vl()` path adds concept alignment.



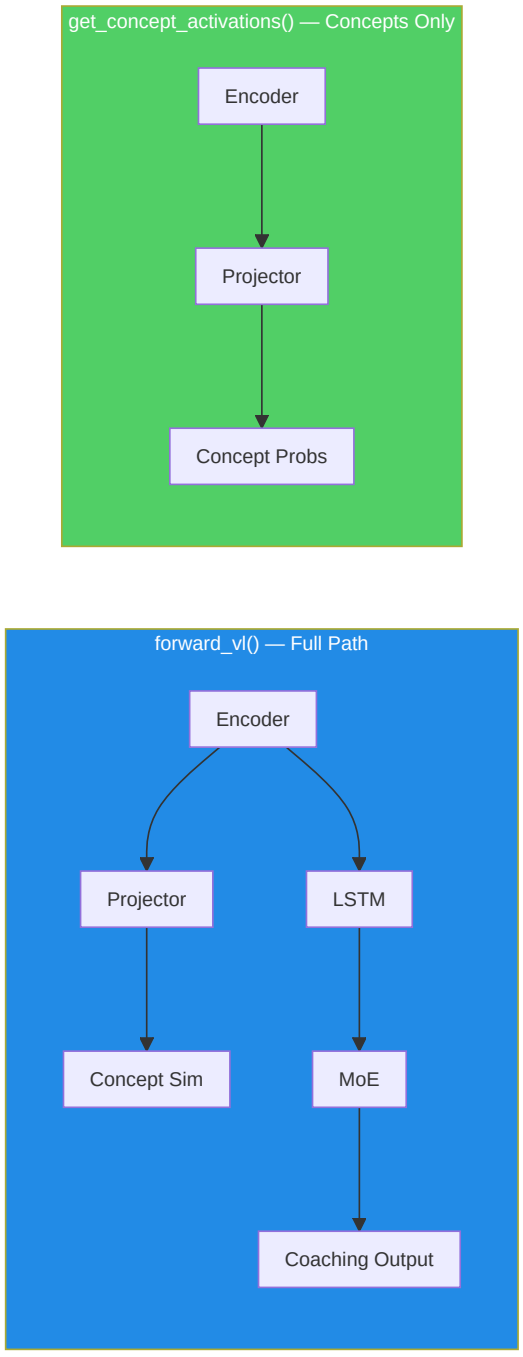
VL-JEPA Forward Path (`forward_vl`)

```
1. Encode:        embeddings = context_encoder(x)                    # [B, seq, 256]
2. Pool:          latent = embeddings.mean(dim=1)                    # [B, 256]
3. Project:       projected = L2_normalize(concept_projector(latent)) # [B, 256]
4. Similarity:    logits = projected @ concept_embs_norm.T            # [B, 16]
5. Temperature:   probs = softmax(logits / clamp(temp, 0.01, 1.0))   # [B, 16]
6. Coaching:      coaching_output = forward_coaching(x, role_id)     # [B, output_dim]
7. Decode:        top_concepts = top-k(probs, k=3)                   # interpretable
```

Output `forward_vl()`: Dictionary with 5 keys:

Key	Shape	Content
`concept_probs`	`[B, 16]`	Softmax probability for each concept
`concept_logits`	`[B, 16]`	Raw similarity scores (pre-softmax)
`top_concepts`	`List[tuple]`	`[(concept_name, probability), ...]` for the first sample
`coaching_output`	`[B, output_dim]`	Standard coaching prediction (via parent)
`latent`	`[B, 256]`	Pooled latent embedding from encoder

Lightweight path — `get_concept_activations()`: Concept-only forward without coaching head or LSTM. Uses `torch.no_grad()` for maximum efficiency during inference.



ConceptLabeler: Two Labeling Modes

The `ConceptLabeler` class generates **soft multi-label labels** (`[0, 1]^16`) for VL-JEPA training. It supports two modes:

Mode 1 — Outcome-based (preferred, G-01 fix): `label_from_round_stats(round_stats)` generates labels from **round outcome** data (kills, deaths, damage, survival, trade kill, utility, equipment, round won). These data are **orthogonal** to the 25-dim input vector, eliminating label leakage.

Concept	Outcome Signal Used
`positioning_aggressive` (0)	`opening_kill=True` → 0.8, kills≥2+survived → 0.6
`positioning_passive` (1)	survived, no opening, damage<60 → 0.7
`positioning_exposed` (2)	`opening_death=True` → 0.8, deaths>0+damage<40 → 0.6
`utility_effective` (3)	utility_total>80 + round_won → 0.5+util/300
`utility_wasteful` (4)	zero utility → 0.5, utility+lost → 0.4
`economy_efficient` (5)	eco win (equip<2000) → 0.9, normal win → 0.7
`economy_wasteful` (6)	high equip+loss → 0.4+equip/16000
`engagement_favorable` (7)	multi-kill+survived → 0.5+kills×0.15
`engagement_unfavorable` (8)	deaths+no kills+low dmg → 0.7
`trade_responsive` (9)	trade_kills>0 → 0.6+tk×0.2
`trade_isolated` (10)	died, not traded, no trade kills → 0.7
`rotation_fast` (11)	assists≥1+round_won → 0.6+assists×0.1
`information_gathered` (12)	flashes≥2+survived → 0.6
`momentum_leveraged` (13)	rating>1.5 → rating/2.5, kills≥3 → 0.7
`clutch_composed` (14)	kills≥2+survived+won → 0.6
`aggression_calibrated` (15)	efficiency = kills×1000/equip → min(eff×0.5, 1.0)

Mode 2 — Legacy heuristic (fallback with label leakage): `label_tick(features)` generates labels directly from the 25-dim feature vector. This creates **label leakage** because the model can "cheat" by reconstructing the input features instead of learning latent patterns. Used only when `RoundStats` is not available, with a one-time log warning.

`label_batch(features_batch)`: Wrapper that handles 2D batches `[B, 25]` and 3D batches `[B, seq_len, 25]` (mean of labels over the sequence for 3D input).

Feature index reference (METADATA_DIM=25):

```
 0: health/100      1: armor/100       2: has_helmet      3: has_defuser
 4: equip/10000     5: is_crouching    6: is_scoped       7: is_blinded
 8: enemies_vis     9: pos_x/4096     10: pos_y/4096     11: pos_z/1024
12: view_x_sin     13: view_x_cos     14: view_y/90      15: z_penalty
16: kast_est       17: map_id         18: round_phase
19: weapon_class   20: time_in_round/115  21: bomb_planted
22: teammates_alive/4  23: enemies_alive/5  24: team_economy/16000
```

VL-JEPA Loss Functions

1. `jepa_contrastive_loss()` — InfoNCE (already documented above)

Formula: `-log(exp(sim(pred, target)/τ) / (exp(sim(pred, target)/τ) + Σ exp(sim(pred, neg_i)/τ)))` with τ=0.07. The PyTorch implementation uses an efficient trick: it concatenates the positive similarity and the negative similarities into a logits vector `[pos_sim, neg_sim_1, ..., neg_sim_N]` and applies `F.cross_entropy(logits, labels=0)` — where label `0` indicates that the first element (the positive similarity) is the correct class. This is mathematically equivalent to the InfoNCE formula but leverages PyTorch's numerically stable internal `log_softmax`.

2. `vl_jepa_concept_loss()` — Concept Alignment + VICReg Diversity

```
concept_loss = BCE_with_logits(concept_logits, concept_labels)  # Multi-label
diversity_loss = -std(L2_normalize(concept_embeddings), dim=0).mean()  # VICReg
total = alpha * concept_loss + beta * diversity_loss
```

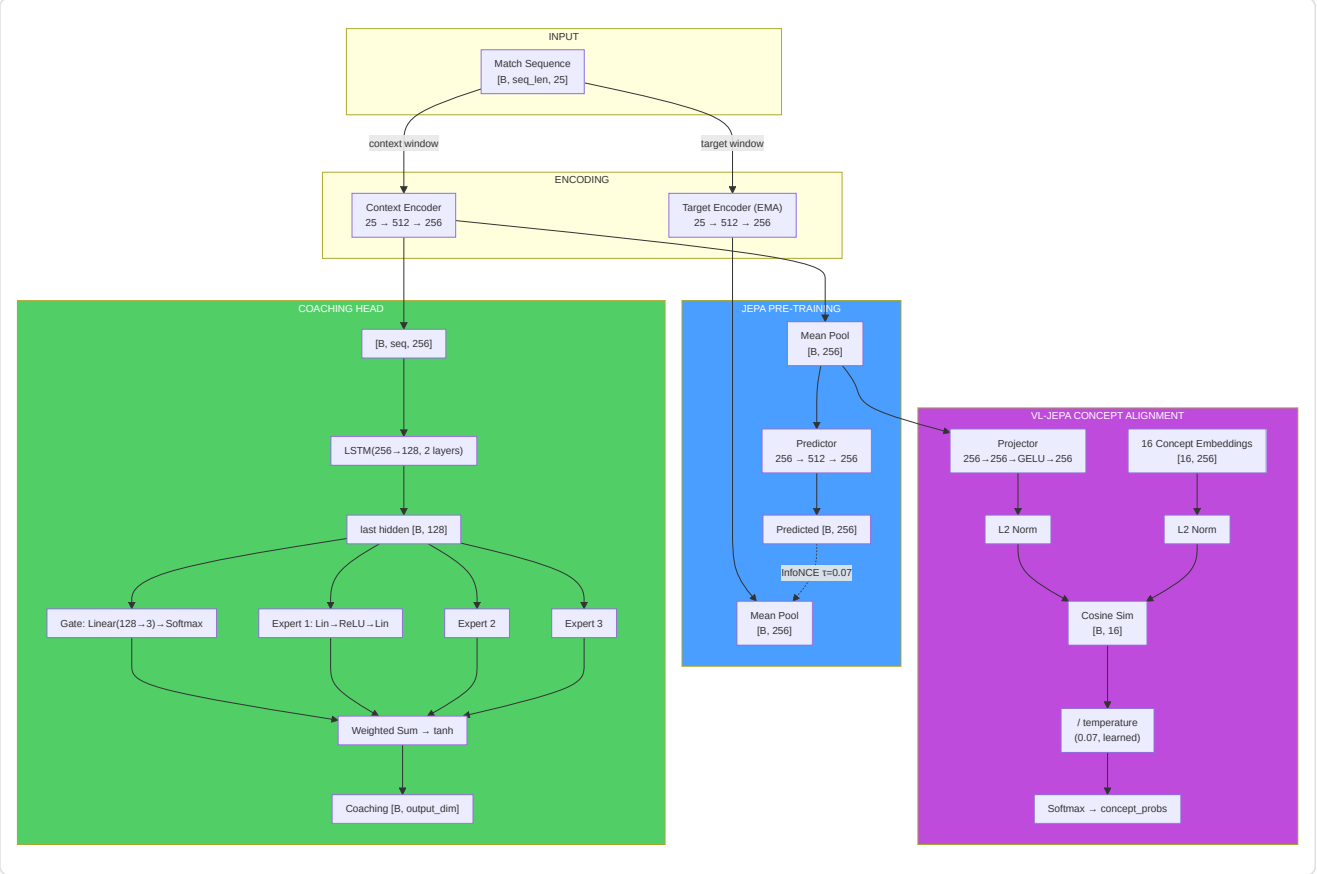
Term	Formula	Default Weight	Purpose
`concept_loss`	`F.binary_cross_entropy_with_logits(logits, labels)`	α = 0.5	Aligns embedding with correct concepts
`diversity_loss`	`-std_per_dim(L2_norm(concept_embs)).mean()`	β = 0.1	Prevents collapse of concept embeddings

Total loss in the VL-JEPA training step (`train_step_vl`):

```
L_total = L_infonce + α × L_concept + β × L_diversity
```

Where `L_infonce` is the standard JEPA contrastive loss and `(α × L_concept + β × L_diversity)` is the concept alignment term.

Complete JEPA / VL-JEPA Dimensional Flow



JEPATrainer: Training with Drift Monitoring

Defined in `jepa_trainer.py`. Manages both standard JEPA training and VL-JEPA, with automatic drift-based retraining.

Parameter	Default	Purpose
Optimizer	AdamW (lr=1e-4, weight_decay=1e-4)	Optimization with weight decay
Scheduler	CosineAnnealingLR (T_max=100)	Cyclic learning rate decay
DriftMonitor	z_threshold=2.5	Detects feature drift beyond 2.5σ
drift_history	`List[DriftReport]`	Drift report history

Shared negative encoding — `encode_raw_negatives(negatives, seq_len)` (NN-H-02):

Shared method that encodes raw negatives (feature space) into the latent space. When the orchestrator sends negatives from the cross-match pool (dimension `METADATA_DIM`), these must be transformed into latent embeddings for contrastive loss computation. The method: reshape `[B*N, 1, D]` → expand by `seq_len` → encode via `target_encoder` with `torch.no_grad()` → mean pool → reshape `[B, N, latent_dim]`. This logic was previously duplicated between trainer and orchestrator; centralization (NN-H-02) prevents misalignments.

Training cycle — `train_step(x_context, x_target, negatives)`:

1. JEPA forward pass: `pred, target = model.forward_jepa_pretrain(context, target)`
2. **Auto-detect raw negatives:** If `negatives.shape[-1] ≠ latent_dim`, the negatives are raw features → they are auto-encoded via `encode_raw_negatives()` (NN-H-02)
3. InfoNCE loss on normalized embeddings
4. Backward + optimizer step
5. **EMA update of target encoder** (must happen AFTER `optimizer.step()`)
6. **Embedding diversity monitoring (P9-02):** Computes the mean variance of latent dimensions. If `variance < 0.01`, emits a warning of potential embedding collapse — the model is converging to a degenerate representation where all inputs produce the same vector

Degenerate batch guard (NN-JT-01): In in-batch negatives mode, if `batch_size < 2` the batch is skipped — it is not possible to build negatives excluding self from a batch of a single element. This prevents errors in negative indexing during the initial training phases with limited data.

VL-JEPA training step — `train_step_vl()`: Extends `train_step` with:

1. Standard JEPA forward pass (InfoNCE)
2. VL forward: `model.forward_vl(x_context)` → concept_logits
3. **Label generation (G-01 preference):** If `round_stats` is available → `label_from_round_stats()` (no leakage). Otherwise → `label_batch()` (legacy heuristic with one-time warning)
4. Concept loss + diversity loss: `vl_jepa_concept_loss(logits, labels, embeddings, α, β)`
5. Total loss: `L_infonce + L_concept_total`
6. Backward + optimize + EMA update

Output: `{total_loss, infonce_loss, concept_loss, diversity_loss}`.

Drift monitoring — `check_val_drift(val_df, reference_stats)`:

- Uses `DriftMonitor` from the validation pipeline
- Computes z-score for each feature of the validation set relative to reference statistics
- If `max_z_score > 2.5`, the report flags `is_drifted=True`
- `should_retrain(drift_history, window=5)` → if the majority of the last 5 windows show drift, triggers the `_needs_full_retrain` flag

Automatic retraining — `retrain_if_needed(full_dataloader, device, epochs=10)`:

- If the `_needs_full_retrain` flag is active, resets the scheduler and reruns `epochs` full epochs
- After retraining, clears the flag and the drift history
- Returns `True/False` to indicate whether retraining occurred

Standalone Training Pipeline (`jepa_train.py`)

Standalone script for JEPA pre-training and fine-tuning, executable from CLI:

```
python -m Programma_CS2_RENAN.backend.nn.jepa_train --mode pretrain
python -m Programma_CS2_RENAN.backend.nn.jepa_train --mode finetune --model-path models/jepa_model.pt
```

`JEPAPretrainDataset`: PyTorch Dataset for pre-training:

Parameter	Default	Description
`context_len`	10	Context window length (ticks)
`target_len`	10	Target window length (ticks)
`match_sequences`	`List[np.ndarray]`	Match sequences `[num_rounds, METADATA_DIM]`

For each sample, it selects a random starting point in the sequence and returns `{"context": [context_len, 25], "target": [target_len, 25]}`.

> **Note (F3-25):** The starting point uses `np.random.randint()` with non-seeded global state → windows not reproducible across runs. For deterministic training, use `worker_init_fn` or a dedicated `Generator` in the `DataLoader`.

`_roundstats_to_features(rs: RoundStats)` → `List[float]`: Extracts a **16-feature** vector from a single `RoundStats` row: `[kills, deaths, damage_dealt/100, headshot_kills, assists, trade_kills, was_traded, opening_kill, opening_death, he_damage/100, molotov_damage/100, flashes_thrown, smokes_thrown, equipment_value/5000, round_rating, side_CT]`. The vector is then padded to `METADATA_DIM` (25) with zeros.

> **Exclusion of `round_won` (P-RSB-03):** The `round_won` field is **deliberately excluded** from the feature vector. Including it would cause data leakage: the model would see the round's outcome in the input data, allowing it to trivially predict outcomes from the result itself. `round_won` is correctly used as a **label** in `label_from_round_stats()` (jepa_model.py), where it generates supervision labels for the VL-JEPA branch.

`_MIN_ROUNDS_FOR_SEQUENCE = 6`: Minimum round requirement to build a valid training sequence. Matches with fewer than 6 rounds do not provide enough temporal context to learn meaningful tactical patterns and are silently discarded.

`load_pro_demo_sequences(limit=100)`: Loads professional demo sequences from the database. Uses `_roundstats_to_features()` to extract real per-round features from `RoundStats`, with a fallback to 12 match-level aggregate features from `PlayerMatchStats` (padded to `METADATA_DIM` with zeros) only when `RoundStats` is not available.

> **Critical Warning (F3-08):** In the match-aggregate fallback path, the script uses `np.tile(features, (20, 1))` to create 20 identical frames from a single aggregated vector. This makes JEPA pre-training **an identity operation** — the model simply learns to copy the input, not temporal dynamics. The primary path with real `RoundStats` and the `TrainingOrchestrator` in the production path **are not affected** by this issue and use real per-round/per-tick data.

`train_jepa_pretrain()`: 50 epochs, batch_size=16, lr=1e-4, 8 in-batch negatives. The optimizer includes ONLY `context_encoder` and `predictor` — the `target_encoder` is updated exclusively via EMA.

`train_jepa_finetune()`: 30 epochs, batch_size=16, lr=1e-3, weight_decay=1e-3. Freezes the encoders and optimizes only LSTM + MoE + Gate.

Persistence: `save_jepa_model()` saves `{model_state_dict, is_pretrained}`. `load_jepa_model()` loads with `weights_only=True` (security).

SuperpositionLayer — Contextual Gating (`layers/superposition.py`)

Standalone module that implements a linear layer with **context-dependent gating**, used within the RAP Coach Strategy Layer.

```
class SuperpositionLayer(nn.Module):
    def __init__(self, in_features, out_features, context_dim=METADATA_DIM):
        self.weight = nn.Parameter(empty(out_features, in_features))
        nn.init.kaiming_uniform_(self.weight, a=math.sqrt(5))  # P1-09: Kaiming init
        self.bias = nn.Parameter(zeros(out_features))
        self.context_gate = nn.Linear(context_dim, out_features)  # Superposition Controller

    def forward(self, x, context):
        gate = sigmoid(self.context_gate(context))  # [B, out_features]
        self._last_gate_live = gate                  # With gradient (for sparsity loss)
        self._last_gate_activations = gate.detach()  # Detached copy (for observability)
        out = F.linear(x, self.weight, self.bias)
        return out * gate  # Contextual modulation
```

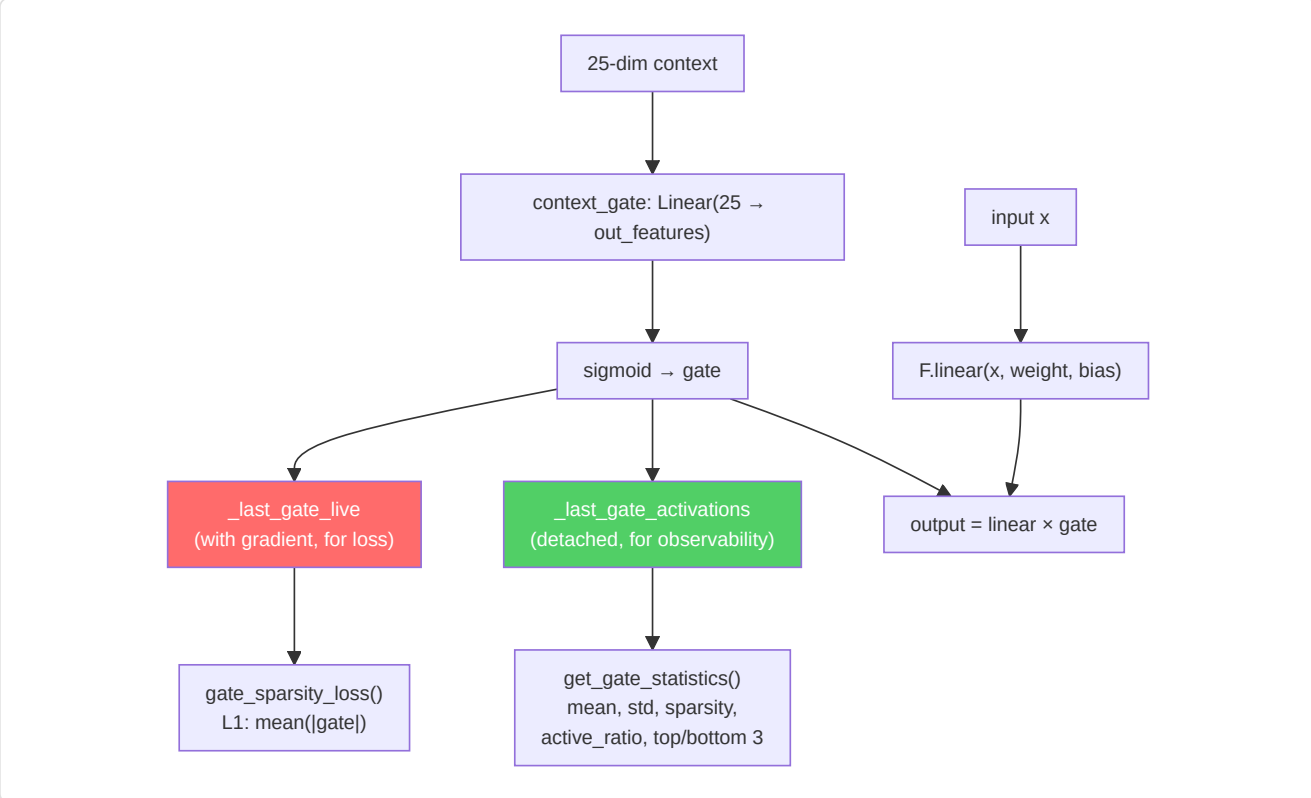
Mechanism: Each neuron's output is multiplied by a sigmoid gate conditioned on context features (25-dim). This allows the model to dynamically "turn on" or "turn off" neurons depending on the game situation.

Kaiming initialization (P1-09): Weights are initialized with `kaiming_uniform_` (Kaiming He distribution, 2015) instead of `torch.randn()`. This initialization ensures that weight variance is proportional to the layer's fan-in, preventing vanishing or exploding gradients in deep networks. The `a=math.sqrt(5)` parameter is the standard value for linear layers in PyTorch.

Dual-tensor design (NN-24): The sigmoid gate is stored in **two separate copies** during each forward pass:

Tensor	Gradient	Purpose
`_last_gate_live`	**Yes** (preserves computational graph)	Used by `gate_sparsity_loss()` for backpropagation — the gradient flows through the gate into `context_gate`
`_last_gate_activations`	**No** (detached)	Used by `get_gate_statistics()` for observability — no memory cost for the graph

This separation resolves the conflict between the need for gradients (for sparsity loss) and the need for lightweight observability (for logging and TensorBoard).



Integrated observability:

Method	Returns	Description
`get_gate_activations()`	`Tensor` or `None`	Latest gate activations (`_last_gate_activations` , detached)
`get_gate_statistics()`	`Dict[str, float]`	Full gate statistics (see table below)
`gate_sparsity_loss()`	`Tensor`	L1 loss `mean(
`enable_tracing(interval)`	—	Detailed gate logging every `interval` steps
`disable_tracing()`	—	Restores logging interval to 100

Fields of `get_gate_statistics()` :

Field	Type	Meaning
`mean_activation`	float	Mean of gate activations in the batch
`std_activation`	float	Standard deviation of activations
`sparsity`	float	Fraction of dimensions with mean < 0.1 (higher = sparser)
`active_ratio`	float	Fraction of dimensions with mean > 0.5 (higher = more active)
`top_3_dims`	List[int]	The 3 most active gate dimensions
`bottom_3_dims`	List[int]	The 3 least active gate dimensions

Periodic logging during training: Every 100 forward passes (configurable via `enable_tracing(interval)`), logs via structured logger: active dimensions (gate_mean > 0.5), sparse dimensions (gate_mean < 0.1), and overall mean.

Standalone EMA Module

The **Exponential Moving Average** update of the target encoder is implemented directly in `JEPACoachingModel.update_target_encoder(momentum=0.996)` :

```
with torch.no_grad():
    for param_q, param_k in zip(context_encoder.parameters(), target_encoder.parameters()):
        param_k.data = param_k.data * momentum + param_q.data * (1.0 - momentum)
```

Invariants:

- The EMA update **always happens after** `optimizer.step()` — never before, otherwise gradients are not yet applied
- The target encoder **never receives direct gradients** — only EMA updates
- The momentum 0.996 means the target encoder "absorbs" only 0.4% of the online encoder's weights at each step — a very conservative update
- `state_dict()` of the model returns **cloned** tensors (`.clone()`) to prevent accidental aliasing — a real bug fixed during audit where `state_dict()` returned direct references to the model tensors instead of copies, causing corruption when the caller modified the dictionary

-CoachTrainingManager (Orchestration)

Defined in `coach_manager.py` . This is the **brain of the training process**, which manages a rigorous **3-level, maturity-based training cycle**, divided into 4 phases:

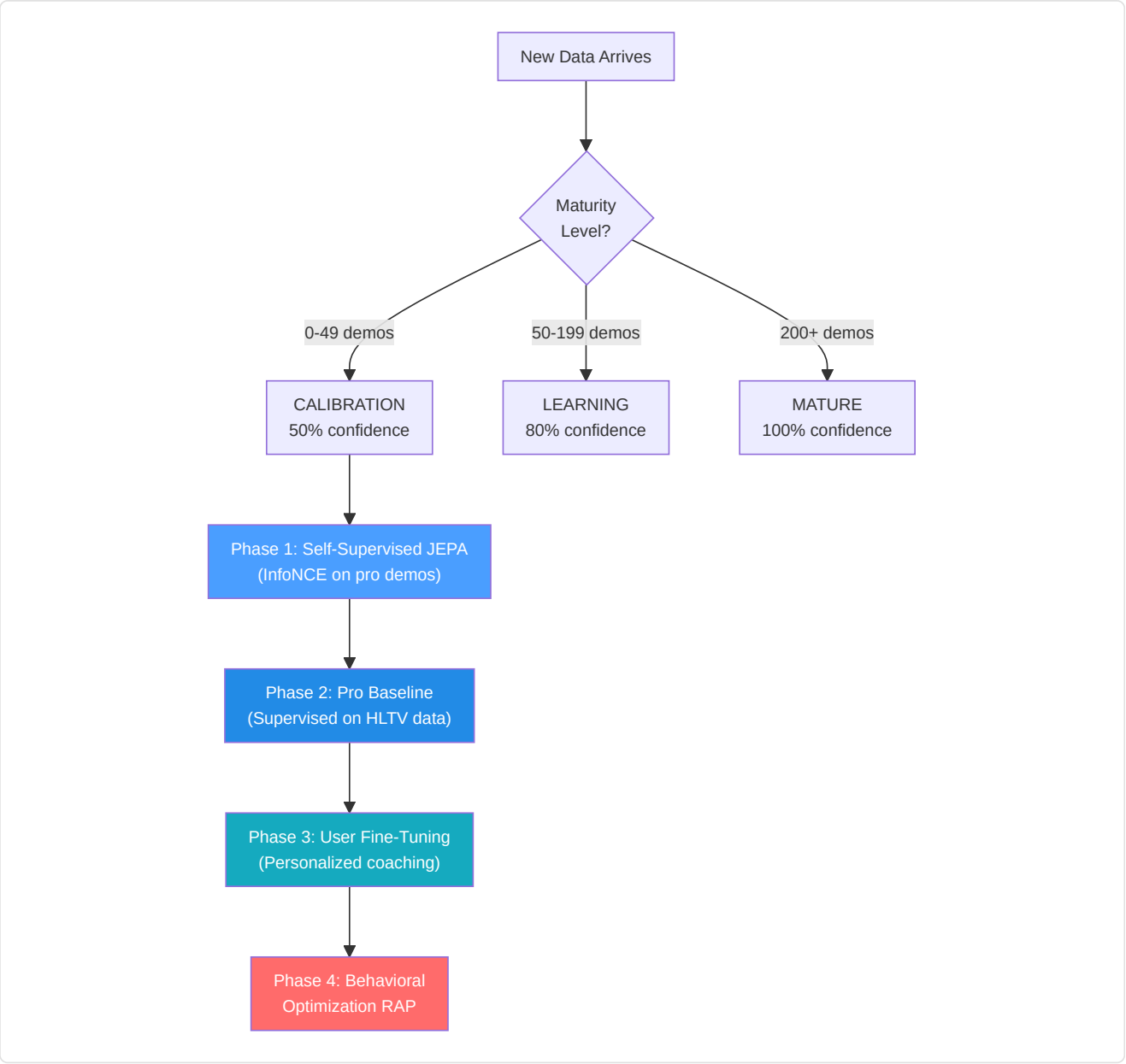
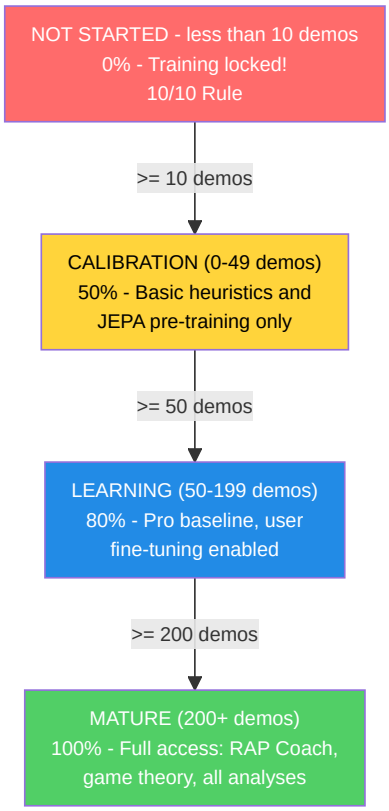


Diagram explanation: Think of the 4 phases as **school years**: Phase 1 (JEPA) is like **watching game film**: the student watches hundreds of pro games and learns the patterns without anyone grading them. Phase 2 (Pro Baseline) is like **studying from a textbook**: now a teacher says "here is how you play well" and the student studies to match. Phase 3 (User fine-tuning) is like **private lessons**: the system adapts specifically to THIS player's style and weaknesses. Phase 4 (RAP) is like an **advanced strategy course**: the full 7-component RAP coach steps in with game theory, positioning, and causal reasoning. You cannot access Phase 4 until Phases 1-3 are complete, just like you cannot study calculus before algebra.

Maturity levels and confidence multipliers:

Level	Demo Count	Confidence Multiplier	Unlocked Features
CALIBRATION	0–49	0.50	Basic heuristics, JEPA pre-training
LEARNING	50–199	0.80	Pro baseline comparison, user fine-tuning
MATURE	200+	1.00	Full RAP Coach, game theory, full analysis



Prerequisites (10/10 Rule): Requires ≥10 professional demos OR (≥10 user demos + connected Steam/FACEIT account) before starting any training.

The manager uses a rigorous **training contract** with 25 features (matching `METADATA_DIM`).

> **Issue resolved (ex G-10):** `coach_manager.py` now defines `TRAINING_FEATURES` with canonical names correct for all 25 indices, perfectly aligned with `vectorizer.py`. The assertion `len(TRAINING_FEATURES) == METADATA_DIM` is valid and all names are up to date. In addition, `MATCH_AGGREGATE_FEATURES` defines the 25 match-level aggregate features: `["avg_kills", "avg_deaths", "avg_adr", "avg_hs", "avg_kast", "kill_std", "adr_std", "kd_ratio", "impact_rounds", "accuracy", "econ_rating", "rating", "opening_duel_win_pct", "clutch_win_pct", "trade_kill_ratio", "flash_assists", "positional_aggression_score", "kpr", "dpr", "rating_impact", "rating_survival", "he_damage_per_round", "smokes_per_round", "unused_utility_per_round", "thrusmoke_kill_pct"]`. Both lists are validated at runtime: if either has a length different from `METADATA_DIM`, the module raises `ValueError` at import time.

```
health, armor, has_helmet, has_defuser, equipment_value,
is_crouching, is_scoped, is_blinded,
enemies_visible,
pos_x, pos_y, pos_z,
view_yaw_sin, view_yaw_cos, view_pitch,
z_penalty, kast_estimate, map_id, round_phase,
weapon_class, time_in_round, bomb_planted,
teammates_alive, enemies_alive, team_economy
```

Target indices: `TARGET_INDICES = list(range(OUTPUT_DIM))` = `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]` — the model predicts improvement deltas for the first **10 match-level aggregate metrics**: `[avg_kills, avg_deaths, avg_adr, avg_hs, avg_kast, kill_std, adr_std, kd_ratio, impact_rounds, accuracy]`.

-TrainingOrchestrator

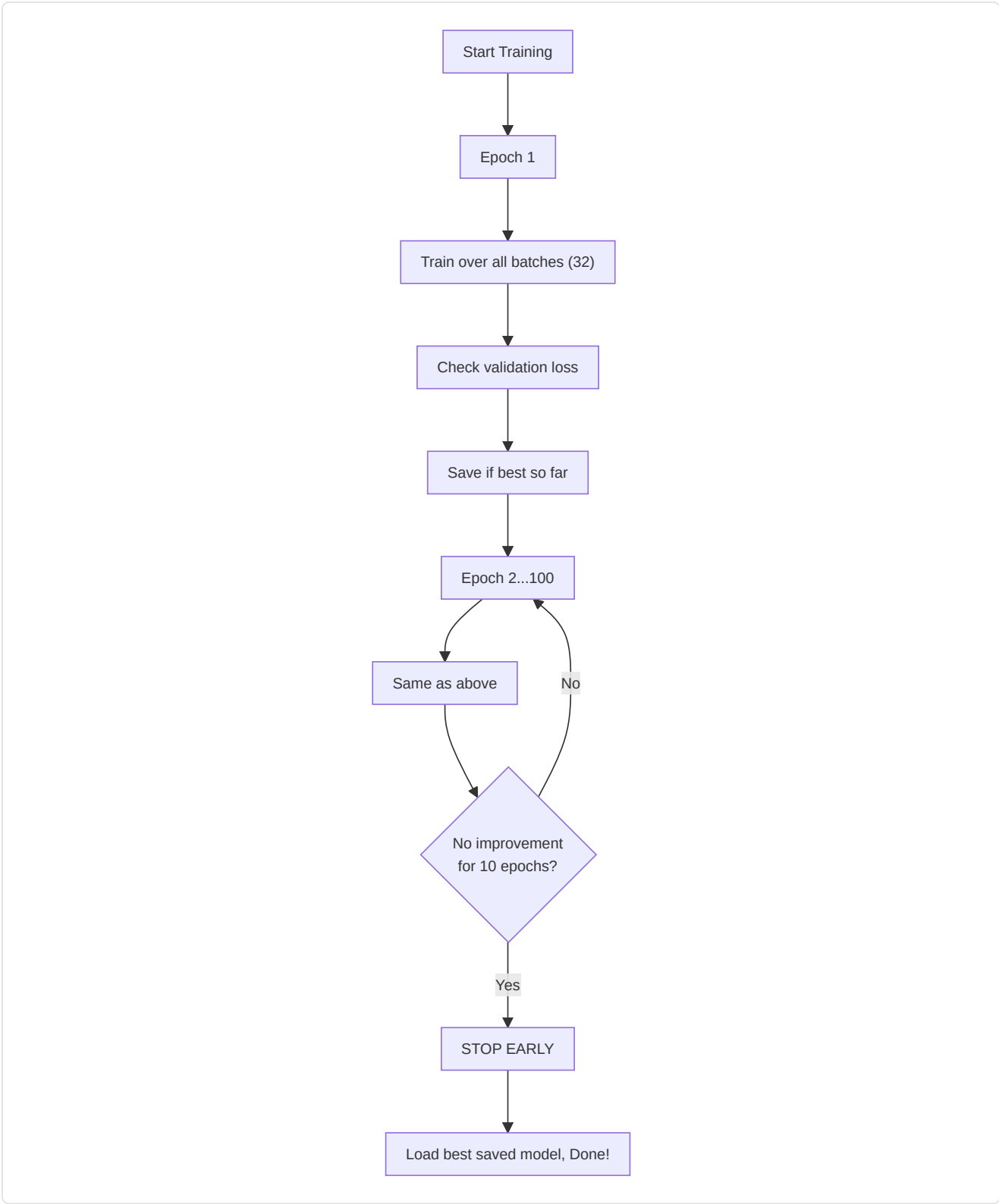
Defined in `training_orchestrator.py`. Unified epoch cycle, validation, early stopping, and checkpoints for JEPA, VL-JEPA, RAP, and RAP Lite models.

Parameter	Default	Purpose
`model_type`	"jepa"	Routes to JEPA, VL-JEPA, RAP, or RAP Lite trainer
`max_epochs`	100	Maximum training limit
`patience`	10	Early stopping patience
`batch_size`	32	Samples per batch
`callbacks`	`None`	List of `TrainingCallback` instances for Observatory integration

The orchestrator integrates with the Observatory via `CallbackRegistry`. It fires lifecycle events at **5 points**: `on_train_start` (before the first epoch), `on_epoch_start` (beginning of each epoch), `on_batch_end` (after each training batch, includes loss output and trainer), `on_epoch_end` (after validation, includes model and losses), `on_train_end` (after training completion or early stopping). When no callbacks are registered, all `fire()` calls are zero-cost operations. Callback errors are caught and logged, never causing the training loop to crash.

Cross-match negative pool (NN-H-03): The orchestrator maintains a pool of feature vectors from previous batches (`_neg_pool`, max 500 vectors). Contrastive negatives are sampled from this pool instead of from the current batch, ensuring negatives come from **different matches** and not from the same temporal context/target sequence. When the pool is still empty (warm-up), the system falls back to in-batch sampling. This avoids false negatives: two ticks from the same gameplay action would be too similar to serve as useful negatives.

Pre-training quality gate (P3-D): Before starting any training, the orchestrator executes `run_pre_training_quality_check()`. If the quality report fails (insufficient data, anomalous distributions, missing features), training is **aborted** with an explanatory error log. This prevents wasting GPU on data that would produce a useless model.



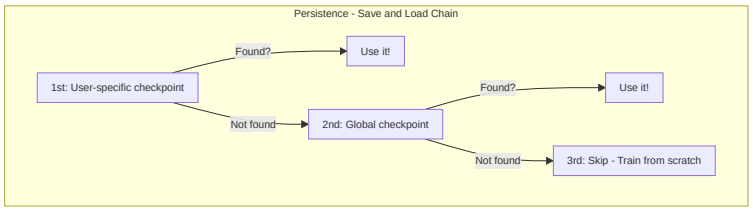
-ModelFactory and Persistence

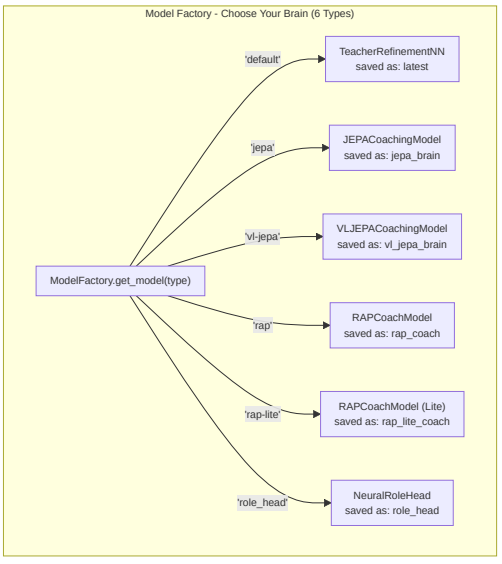
ModelFactory (`factory.py`) provides unified model instantiation:

Type Constant	Model Class	Checkpoint Name	Factory Defaults
`TYPE_LEGACY` ("default")	`TeacherRefinementNN`	`"latest"`	`input_dim=METADATA_DIM(25)`, `output_dim=OUTPUT_DIM(10)`, `hidden_dim=HIDDEN_DIM(128)`
`TYPE_JEPA` ("jepa")	`JEPACoachingModel`	`"jepa_brain"`	`input_dim=METADATA_DIM(25)`, `output_dim=OUTPUT_DIM(10)`
`TYPE_VL_JEPA` ("vl-jepa")	`VLJEPACoachingModel`	`"vl_jepa_brain"`	`input_dim=METADATA_DIM(25)`, `output_dim=OUTPUT_DIM(10)`
`TYPE_RAP` ("rap")	`RAPCoachModel`	`"rap_coach"`	`metadata_dim=METADATA_DIM(25)`, `output_dim=10`
`TYPE_RAP_LITE` ("rap-lite")	`RAPCoachModel`	`"rap_lite_coach"`	`metadata_dim=METADATA_DIM(25)`, `output_dim=10`, `use_lite_memory=True`
`TYPE_ROLE_HEAD` ("role_head")	`NeuralRoleHead`	`"role_head"`	`input_dim=5`, `hidden_dim=32`, `output_dim=5`

> **Note (P1-08):** In a previous version, the factory used `output_dim=4` and `hidden_dim=64` for legacy models, creating a mismatch with `CoachNNConfig`. This has been fixed: now all coaching models (Legacy, JEPA, VL-JEPA) use `OUTPUT_DIM = 10` — the first 10 core aggregate features on which the model predicts delta adjustments. `HIDDEN_DIM = 128` is aligned in both `config.py` and `factory.py`. The RAP and RAP Lite models share the same `RAPCoachModel` class, but RAP Lite activates `use_lite_memory=True`, replacing LTC-Hopfield memory with a pure-PyTorch LSTM fallback (useful when `ncps` / `hflayers` dependencies are not available). The RAP model is imported from the canonical path `backend/nn/experimental/rap_coach/model.py` (the old `backend/nn/rap_coach/model.py` is a redirection shim).
>
> **StaleCheckpointError:** If the dimensions of a saved checkpoint do not match the current model configuration (for example after an upgrade from `output_dim=4` to `output_dim=10`), the system raises `StaleCheckpointError` instead of silently loading incompatible weights, preventing silent corruption.

Persistence (`persistence.py`): Save/load with `weights_only=True` (security), graceful fallback chain (user-specific → global → skip), mismatched size handling.





-Configuration (`config.py`)

```
GLOBAL_SEED = 42                    # Global reproducibility (AR-6, P1-02)
INPUT_DIM = METADATA_DIM = 25       # Canonical 25-dim vector (was 19, was legacy 12)
OUTPUT_DIM = 10                     # First 10 core features the model predicts adjustments for
HIDDEN_DIM = 128                    # Hidden size for AdvancedCoachNN / TeacherRefinementNN
BATCH_SIZE = 32
LEARNING_RATE = 0.001
EPOCHS = 50
RAP_POSITION_SCALE = 500.0          # P9-01: Scale factor for position delta ([-1,1] → world units)
```

> **Note:** `INPUT_DIM` is imported from `feature_engineering/__init__.py` where `METADATA_DIM = 25`. `OUTPUT_DIM = 10` defines the number of features on which the model produces adjustment predictions — the first 10 match aggregate features (avg_kills, avg_deaths, avg_adr, avg_hs, avg_kast, kill_std, adr_std, kd_ratio, impact_rounds, accuracy). This design choice focuses the model's predictive capacity on the most actionable performance metrics, rather than trying to predict all 25 features (many of which are contextual and not directly improvable by the player). `RAP_POSITION_SCALE = 500.0` is the canonical factor to convert the RAP model's normalized outputs (in the [-1, 1] range) into displacements in CS2 world units.
>
> **Architectural note:** The `CoachNNConfig` dataclass in `model.py` defines `output_dim = METADATA_DIM` (25) as default, but `ModelFactory` always overrides this value with `OUTPUT_DIM = 10` during instantiation. The effective output_dim in production for all models (Legacy, JEPA, VL-JEPA, RAP, RAP Lite) is therefore **10**, not 25. Historically, `OUTPUT_DIM` was 4 (4 selected metrics), then raised to 10 to cover the most relevant aggregate features.

Device management: `get_device()` implements a **3-tier smart GPU selection**:

1. **User override:** If `CUDA_DEVICE` is configured (e.g. "cuda:0" or "cpu"), use it
2. **Automatic discrete GPU:** `_select_best_cuda_device()` enumerates all CUDA devices and selects the one with the most VRAM, **penalizing integrated GPUs** (Intel UHD, Iris) via keyword matching. On multi-GPU systems (e.g. Intel UHD + NVIDIA GTX 1650), the discrete GPU always wins
3. **CPU fallback:** If no CUDA GPU is available

Batch sizing based on ML intensity: `High=128`, `Medium=32`, `Low=8`. The throttling delay between batches adapts: `High=0.0s`, `Medium=0.05s`, `Low=0.2s`.

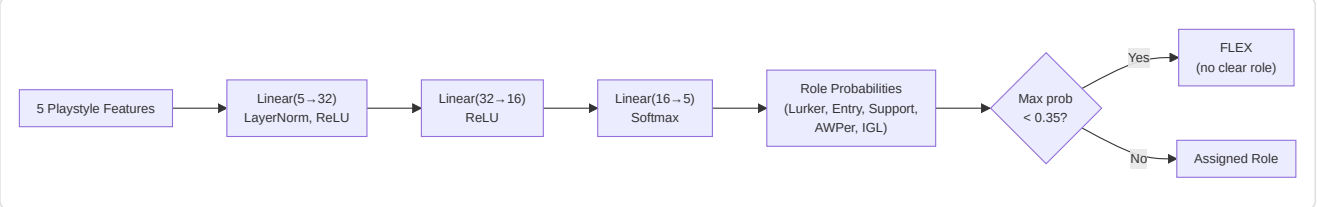
-NeuralRoleHead (MLP for Role Classification)

Defined in `role_head.py`. A lightweight MLP that predicts player role probabilities based on 5 playstyle parameters, operating as a **second opinion** alongside the heuristic `RoleClassifier`. The consensus logic in `role_classifier.py` merges both opinions to produce the final classification.

Architecture:

```
Input (5 features) → Linear(5, 32) → LayerNorm(32) → ReLU
→ Linear(32, 16) → ReLU
→ Linear(16, 5) → Softmax → 5 role probabilities
```

~750 learnable parameters. Minimal compute cost, suitable for per-match inference.



Input features (5 dimensions):

#	Feature	Source	Range	Meaning
0	TAPD	`rounds_survived / rounds_played`	[0, 1]	Survival rate — higher = more passive/support
1	OAP	`entry_frags / rounds_played`	[0, 1]	Opening aggression — higher = entry fragger
2	PODT	`was_traded_ratio`	[0, 1]	Traded-death percentage — higher = traded/baited
3	rating_impact	`impact_rating` or HLTV 2.0	float	Overall round impact
4	aggression_score	`positional_aggression_score`	float	Tendency for advanced positioning

Output roles (5-dim softmax):

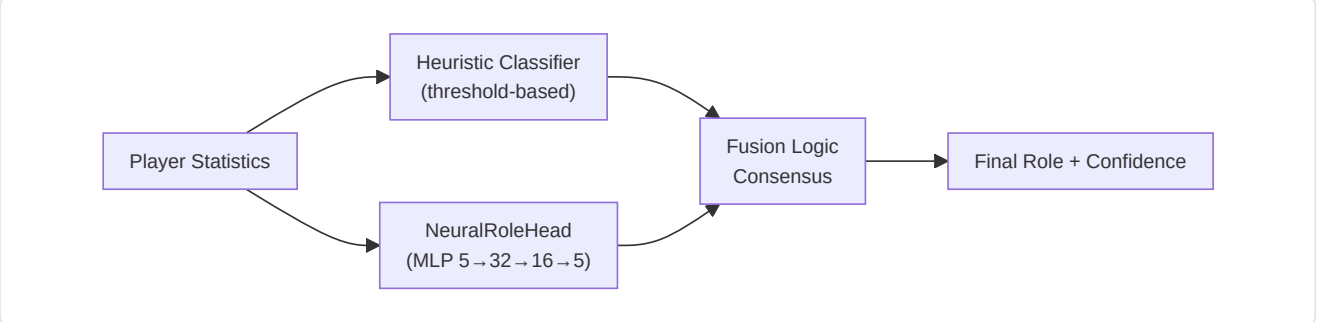
Index	Role	Description
0	LURKER	Hides behind enemy lines
1	ENTRY_FRAGGER	First to enter, takes opening duels
2	SUPPORT	Site anchor, utility usage, trades
3	AWPER	Sniper specialist
4	IGL	In-game leader, tactical director

FLEX threshold: If `max(probabilities) < 0.35`, the player is classified as **FLEX** (versatile, no dominant role). This prevents the model from forcing a role when the player is really a generalist.

Training details:

Aspect	Value
Loss	`KLDivLoss(reduction="batchmean")` on log-softmax predictions against soft label targets
Label smoothing	ε = 0.02 (prevents log(0), adds regularization)
Optimizer	AdamW (lr=1e-3, weight_decay=1e-4)
Early stopping	Patience = 15 epochs on validation loss
Max epochs	200
Train/Val split	80/20 random (cross-sectional data, not sequential)
Minimum samples	20 (from the `Ext_PlayerPlaystyle` table)
Data source	`cs2_playstyle_roles_2024.csv` → `Ext_PlayerPlaystyle` DB table
Normalization	Mean/std per feature computed at training time, saved to `role_head_norm.json`

Consensus with heuristic classifier (`role_classifier.py`):



- **Both agree** → confidence boosted by +0.10

- **Disagree, neural margin > 0.1** → neural opinion wins
- **Disagree, neural margin ≤ 0.1** → heuristic opinion wins
- **Neural unavailable** (no checkpoint or norm_stats) → heuristic only
- **Cold-start protection** → returns FLEX with 0% confidence if thresholds have not been learned

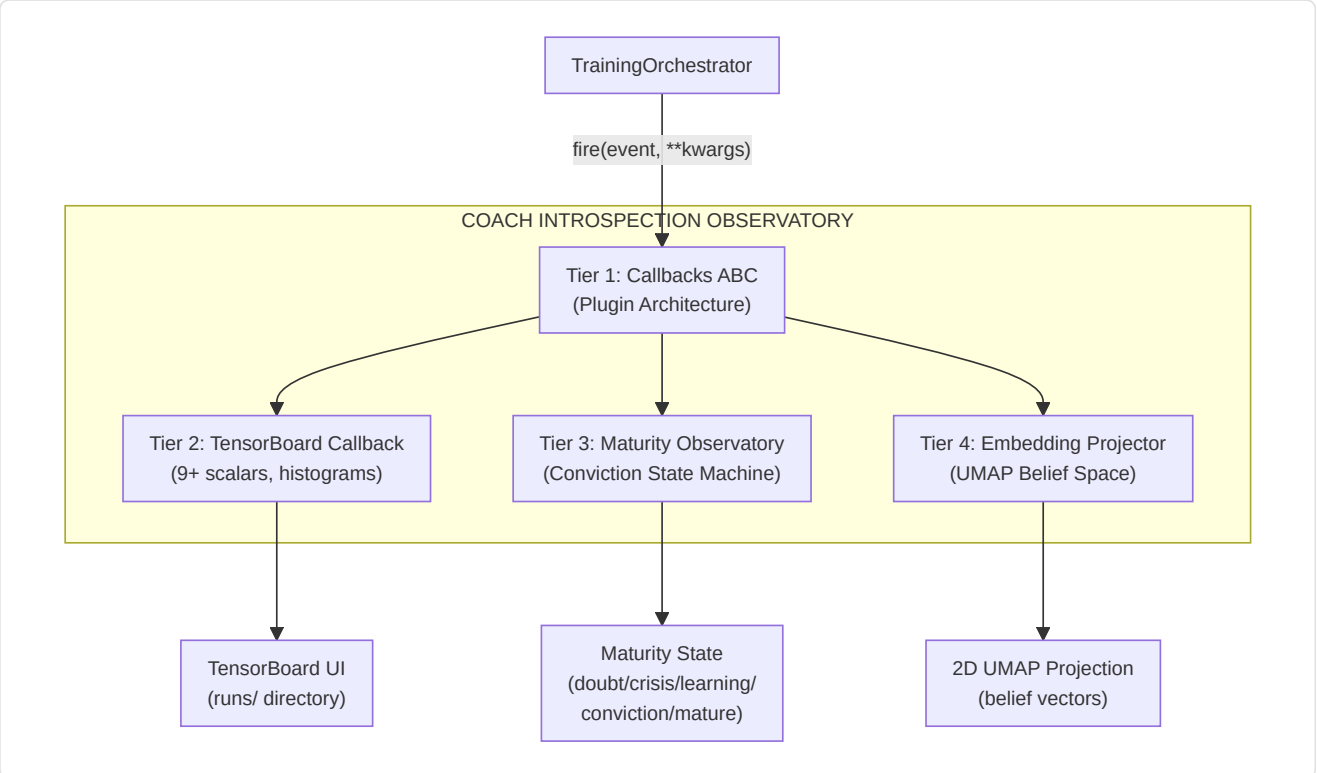
-Coach Introspection Observatory

Files: `training_callbacks.py`, `tensorboard_callback.py`, `maturity_observatory.py`, `embedding_projector.py`

The Observatory is a **4-tier plugin architecture** that instruments the training loop without modifying the core training code. It monitors the coach's neural signals during training and translates them into human-interpretable maturity states, enabling developers and operators to understand whether the model is confused, learning, or production-ready.

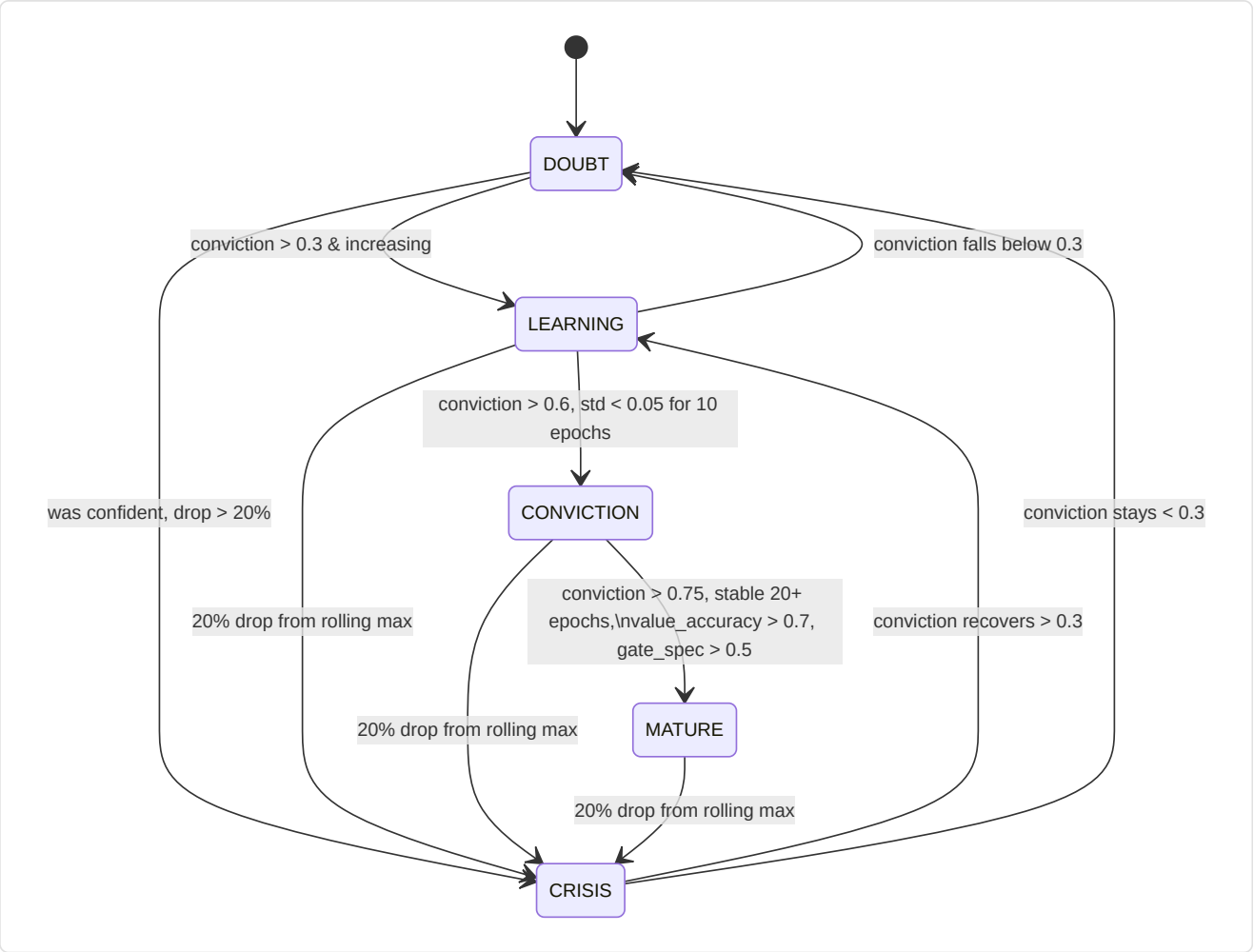
4-tier architecture:

Tier	File	Purpose	Key Output
1. **Callback ABC**	`training_callbacks.py`	Plugin interface + dispatch registry	`TrainingCallback` ABC, `CallbackRegistry.fire()`
2. **TensorBoard**	`tensorboard_callback.py`	Scalar + histogram logging	9+ scalar signals, parameter/grad histograms, gate/belief/concept histograms
3. **Maturity**	`maturity_observatory.py`	Conviction state machine	5 signals → `conviction_index` → 5 maturity states
4. **Embedding**	`embedding_projector.py`	UMAP belief/concept projection	Interactive 2D UMAP figures (graceful degradation if umap-learn is not installed)



Maturity State Machine:

The `MaturityObservatory` computes a **composite conviction index** from 5 neural signals, smooths it with EMA (α=0.3), and classifies the model into one of 5 maturity states:



5 Maturity Signals:

Signal	Weight	Range	What It Measures	Source
`belief_entropy`	0.25	[0, 1]	Shannon entropy of the 64-dim belief vector (lower = more confident)	`model._last_belief_batch`
`gate_specialization`	0.25	[0, 1]	`1 - mean_gate_activation` (higher = more specialized experts)	`SuperpositionLayer.get_gate_statistics()`
`concept_focus`	0.20	[0, 1]	`1 - entropy(concept_embedding_norms)` (lower entropy = focused)	`model.concept_embeddings`
`value_accuracy`	0.20	[0, 1]	`1 - (val_loss / initial_val_loss)` (higher = better calibration)	Validation loop
`role_stability`	0.10	[0, 1]	Conviction consistency across recent epochs (`1 - std*5`)	Self-referential history

Conviction formula:

```
conviction_index = 0.25 × (1 - belief_entropy)
+ 0.25 × gate_specialization
+ 0.20 × concept_focus
+ 0.20 × value_accuracy
+ 0.10 × role_stability

maturity_score = EMA(conviction_index, α=0.3)
```

State thresholds:

State	Condition
DOUBT	`conviction < 0.3`
CRISIS	`conviction` drops > 20% from rolling maximum within 5 epochs
LEARNING	`conviction ∈ [0.3, 0.6]` and increasing
CONVICTION	`conviction > 0.6`, stable (`std < 0.05` over 10 epochs)
MATURE	`conviction > 0.75`, stable for 20+ epochs, `value_accuracy > 0.7`, `gate_specialization > 0.5`

`MaturitySnapshot` **dataclass:** Every epoch, the Observatory produces an immutable snapshot:

Field	Type	Description
`epoch`	int	Epoch number
`timestamp`	datetime	Time of recording
`belief_entropy`	float	Shannon entropy of the belief vector
`gate_specialization`	float	Expert specialization
`concept_focus`	float	Focus on coaching concepts
`value_accuracy`	float	Value prediction accuracy
`role_stability`	float	Role classification stability
`conviction_index`	float	Weighted composite index
`maturity_score`	float	EMA-smoothed score (α=0.3)
`state`	str	Current state (DOUBT/CRISIS/LEARNING/CONVICTION/MATURE)

Extraction of the 5 neural signals — how they are computed:

Signal	Method	Data Source	Computation
`belief_entropy`	`_compute_belief_entropy()`	`model._last_belief_batch`	softmax(belief) → Shannon entropy / log(dim) → 1 - normalized
`gate_specialization`	`_compute_gate_specialization()`	`strategy.superposition.get_gate_statistics()`	`1 - mean_activation` (higher = more specialized experts)
`concept_focus`	`_compute_concept_focus()`	`model.concept_embeddings.weight`	L2 norms → softmax → `1 - entropy` (lower = more focused)
`value_accuracy`	`_compute_value_accuracy()`	Validation cycle	`1 - (val_loss / initial_val_loss)`, clamped [0, 1]
`role_stability`	`_compute_role_stability()`	Recent history	`1 - std(last 10 conviction_index) × 5`, clamped [0, 1]

Public API: `current_state` (property → state string), `current_conviction` (property → float), `get_timeline()` (→ list of `MaturitySnapshot` for export/plots).

TensorBoard integration: Each `on_epoch_end()` logs **7 scalars** to TensorBoard:

```
maturity/belief_entropy, maturity/gate_specialization, maturity/concept_focus,
maturity/value_accuracy, maturity/role_stability,
maturity/conviction_index, maturity/maturity_score
```

Plus a text log of the current state via structured logger.

Design guarantees:

- **Zero impact if disabled:** When no callbacks are registered, all `CallbackRegistry.fire()` calls are no-ops. No memory allocation, no compute overhead.
- **Error isolation:** Each callback is individually try/except-wrapped. A TensorBoard write error or a UMAP computation error never causes the training loop to crash: the error is logged and training continues.
- **Composable:** New callbacks can be added by subclassing `TrainingCallback` and registering with `CallbackRegistry.add()`. No need to modify the training code.

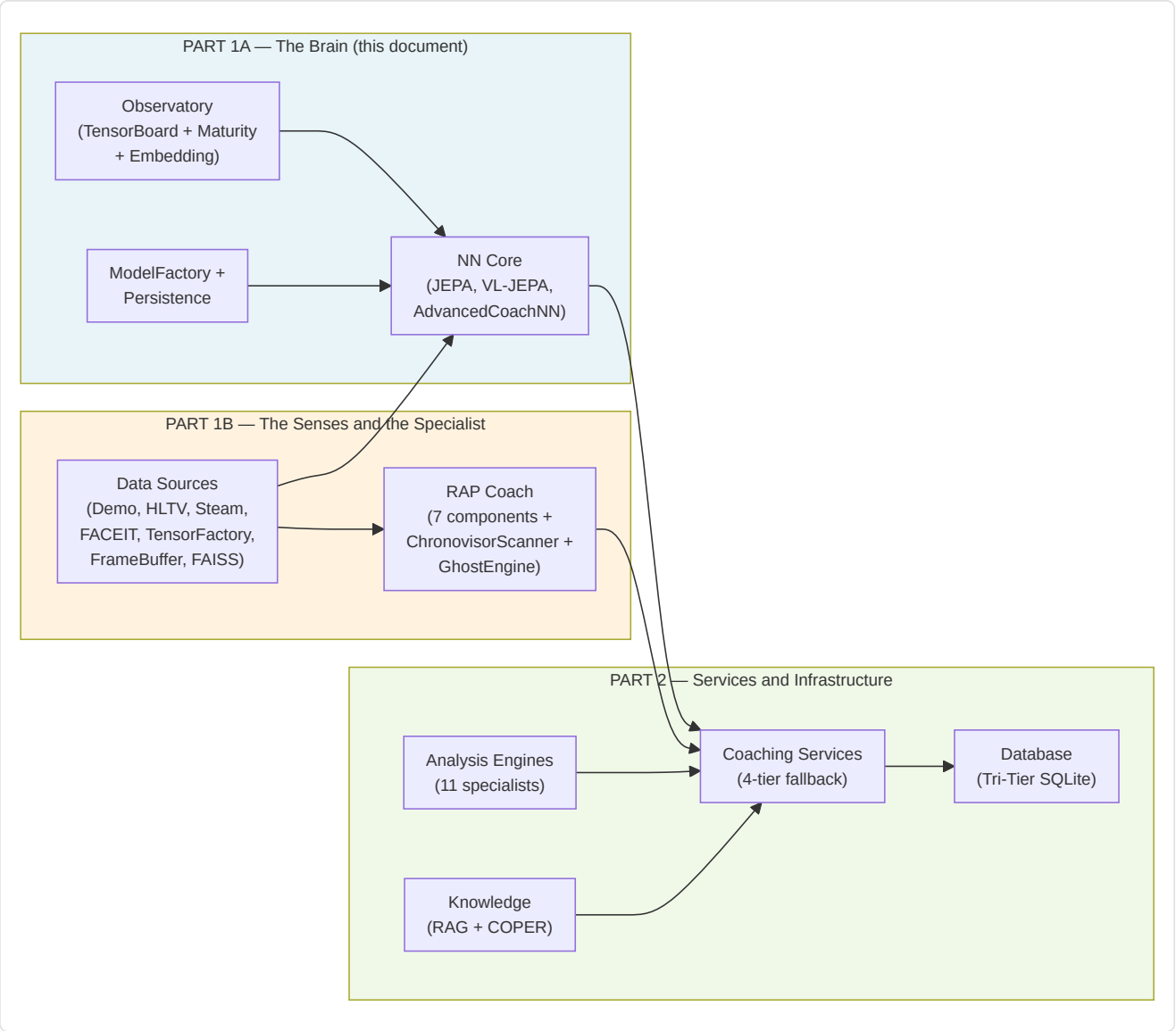
CLI integration: Started via `run_full_training_cycle.py` with the flags:

- `--no-tensorboard` — disables the TensorBoard callback
- `--tb-logdir <path>` — sets the TensorBoard log directory (default: `runs/`)
- `--umap-interval <N>` — UMAP projection every N epochs (default: 10)

Part 1A Summary — The Brain

Part 1A documented the **cognitive core** of the coaching system — the entire neural network subsystem that constitutes the coach's "brain":

Component	Role	Key Details
AdvancedCoachNN	Supervised baseline coaching	2-layer LSTM + 3 MoE experts, 25-dim input, 10-dim output
JEPA	Self-supervised pre-training	Online/target encoder (EMA τ=0.996), InfoNCE contrastive
VL-JEPA	Vision-language alignment	16 coaching concepts across 5 tactical dimensions
SuperpositionLayer	Contextual gating	25-dim context-dependent modulation with integrated observability
CoachTrainingManager	Training orchestration	3 maturity levels (CALIBRATION→LEARNING→MATURE)
TrainingOrchestrator	Unified epoch cycle	Early stopping, checkpoints, callbacks, cross-match negative pool, quality gate
ModelFactory	Model instantiation	6 model types (+ RAP Lite) with persistence and fallback
NeuralRoleHead	Role classification	MLP 5→32→16→5, consensus with heuristic
MaturityObservatory	Training introspection	5 signals → conviction index → 5 maturity states



Continued in Part 1B — *The Senses and the Specialist: RAP Coach Model, ChronovisorScanner, GhostEngine, Data Sources (Demo Parser, HLTV, Steam, FACEIT, TensorFactory, FrameBuffer, FAISS, Round Context)*